

Introduction to electronics

Day #1

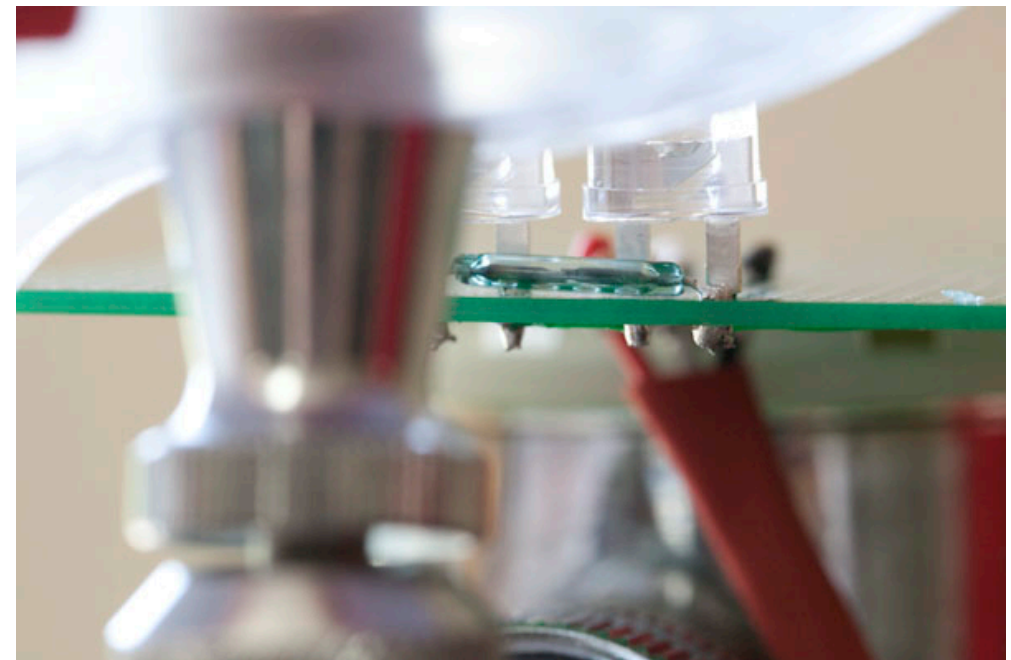
What are we going to learn?

- How to control anything that is mechanical or electrical!
- Who cares?
- What can I do with this?

So, how much (root) beer is left in my keg?



100-lb Force sensor reports weight



Switch closes when a magnetic
field is near

Keg monitor

How can we make mowing the lawn fun?



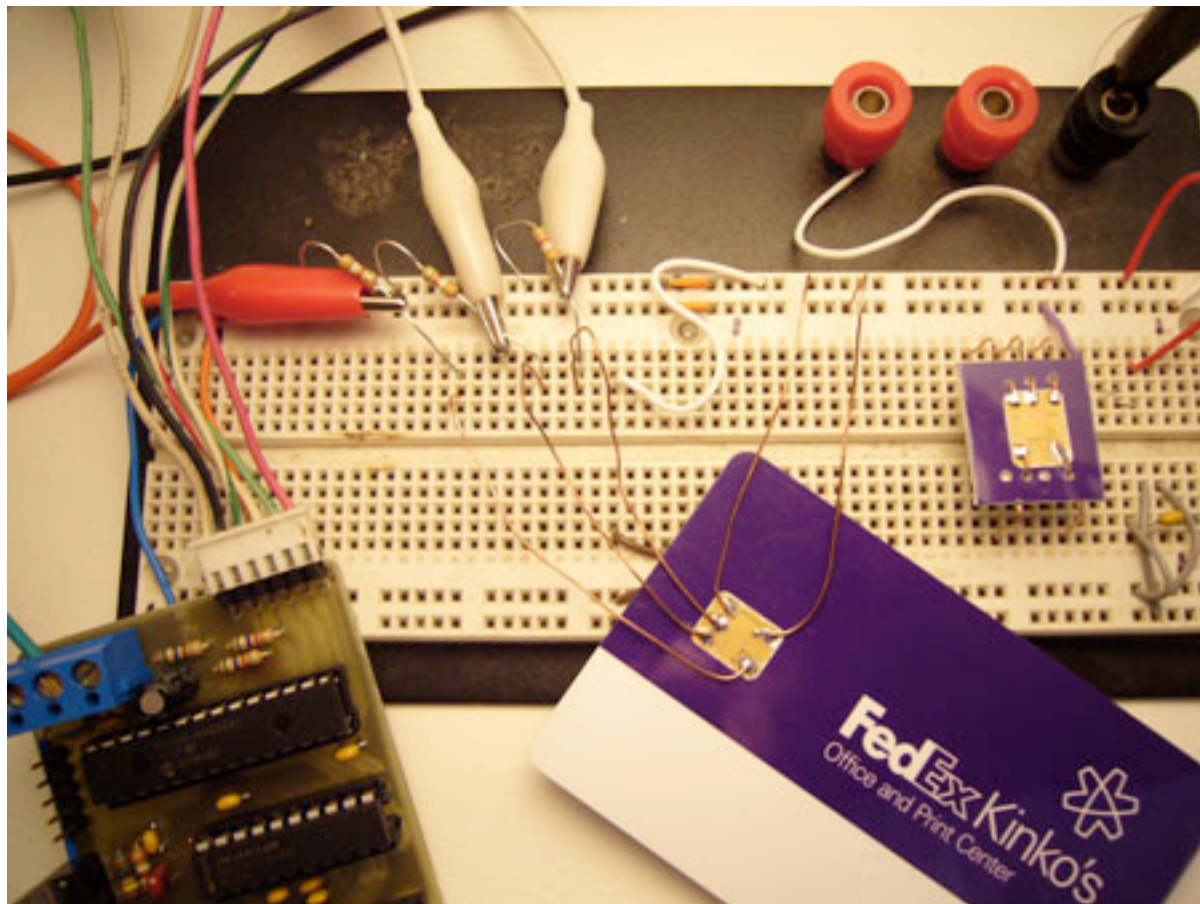
Mowing the lawn - [Video](#)



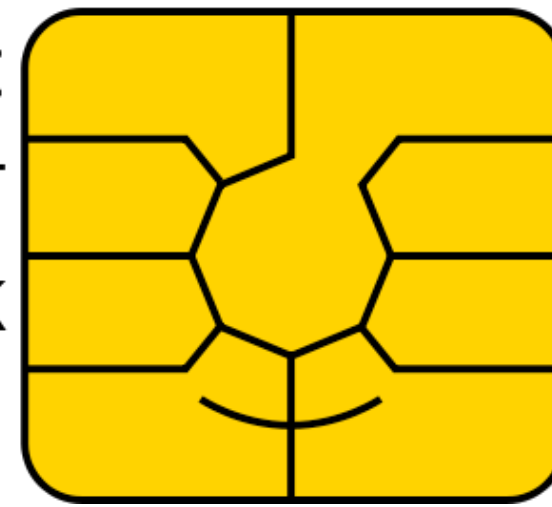
Carrying topsoil - [Video](#)

Remote controlled
lawnmower

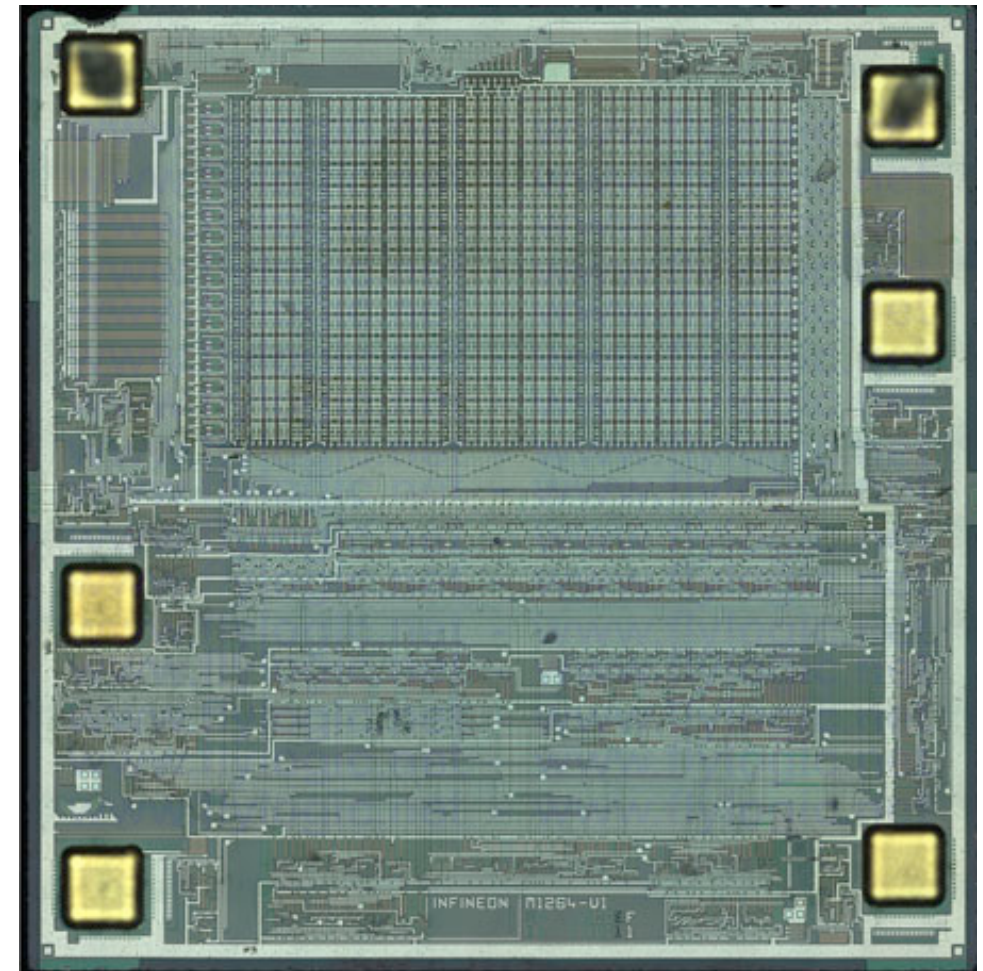
How much money I have left?



C1—VCC
C2—RST
C3—CLK
C4—



C5—GND
C6—VPP
C7—I/O
C8—

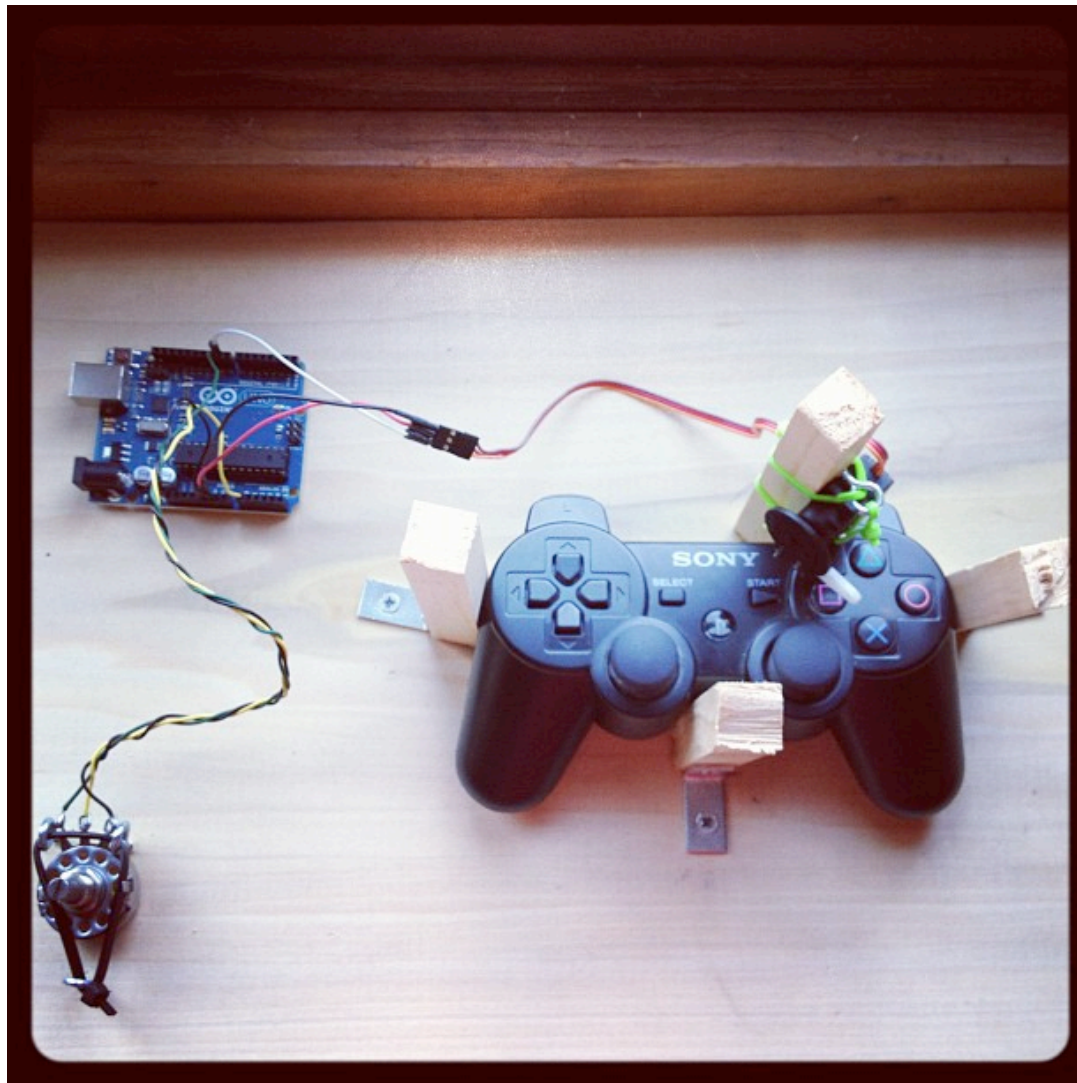


Reading smart cards

What more fun that playing Castlevania?

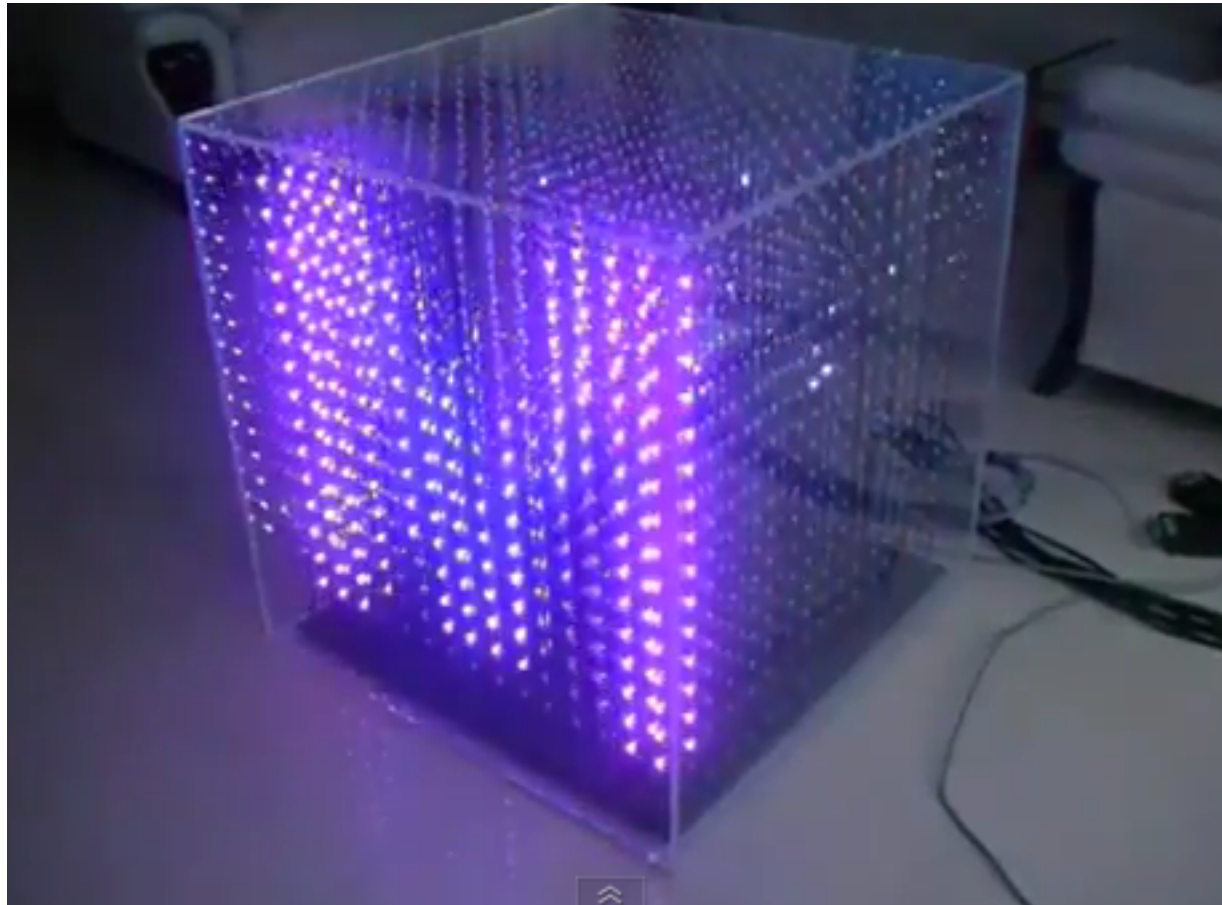


Grinding - Video

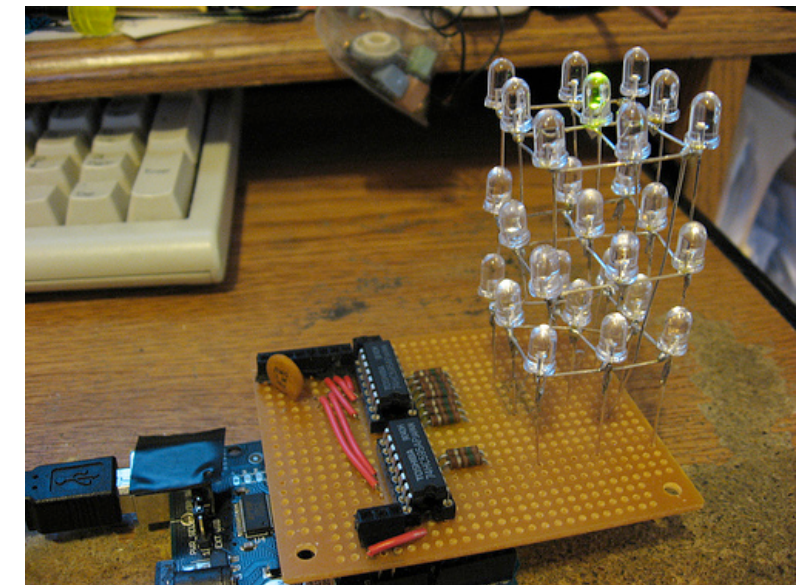


Automated gameplay

You can create your own 3D display.



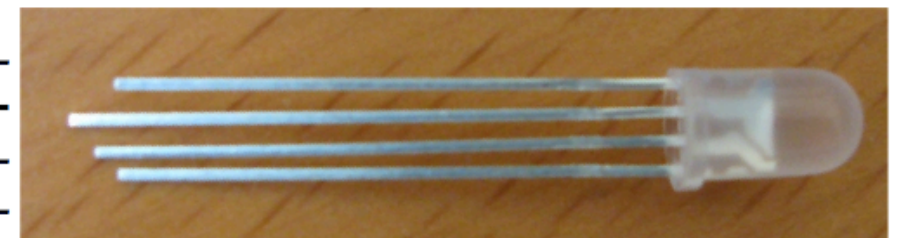
In action - [Video](#)



anode +
cathode -



red cathode -
anode +
green cathode -
blue cathode -



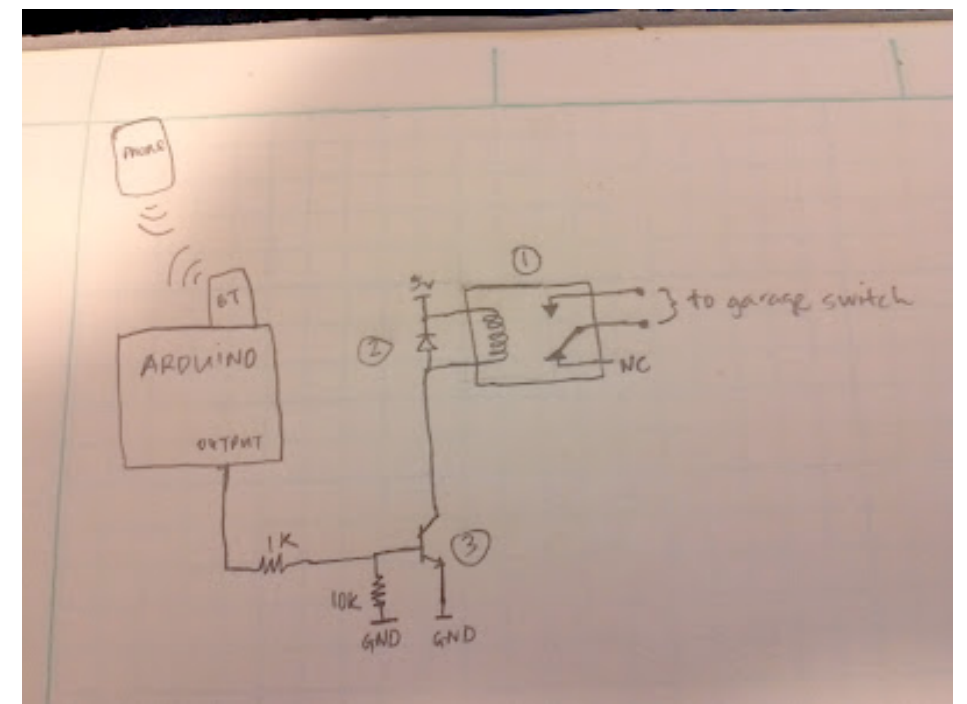
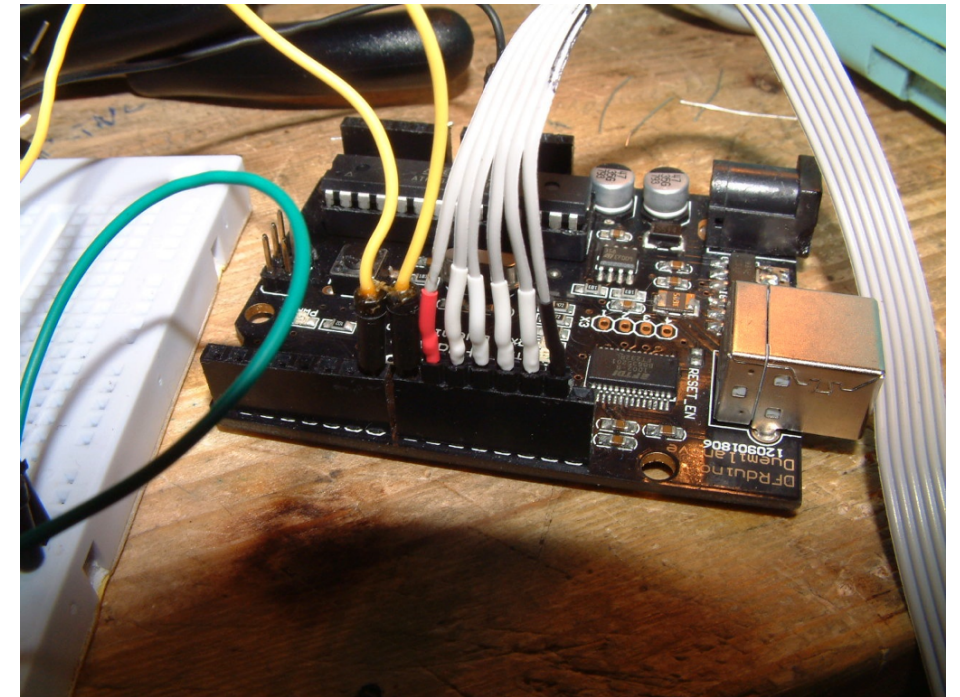
3D RGB LED Cube

I want an app to open my garage door!



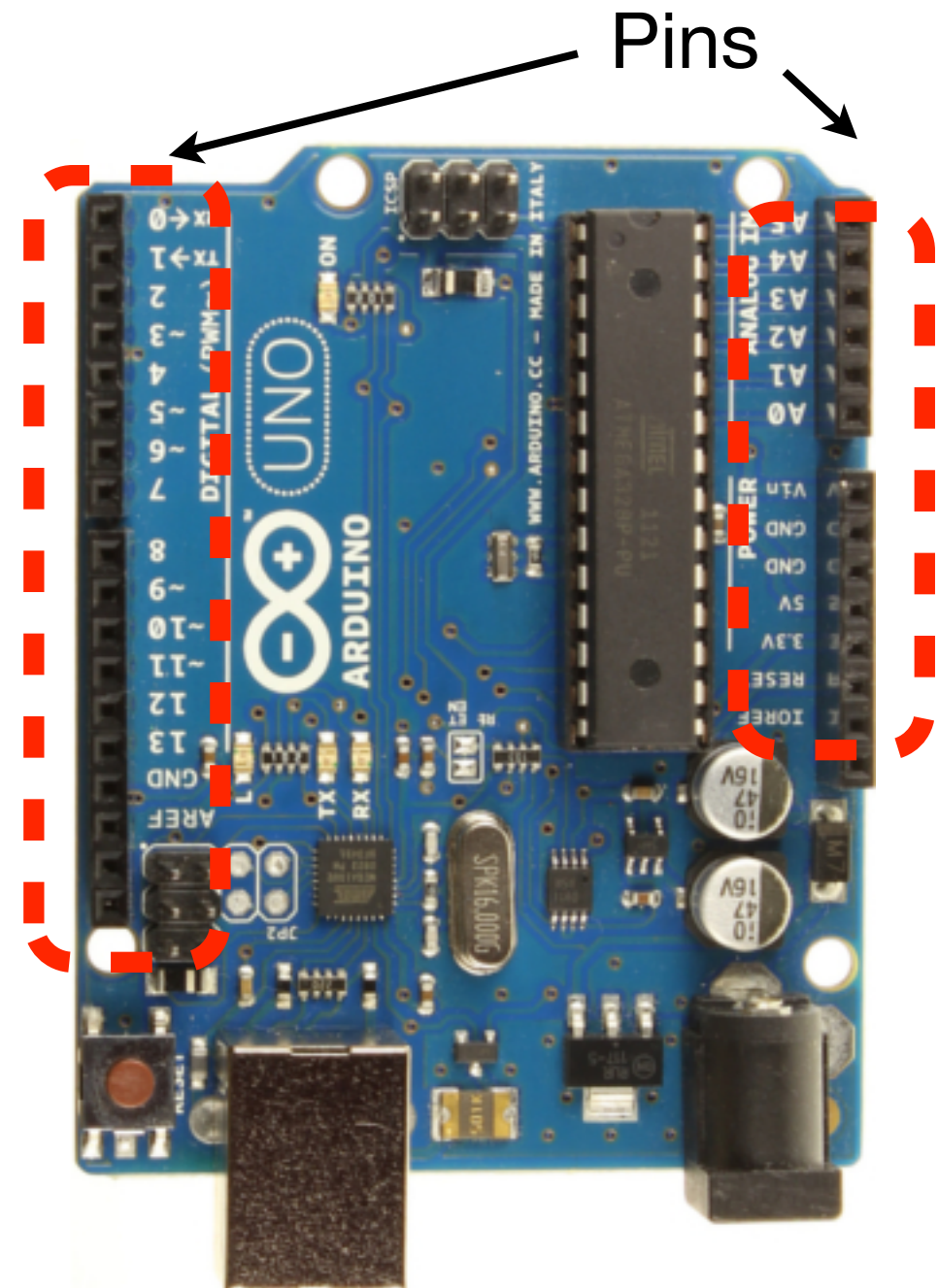
In action - [Video](#)

Garage door opener



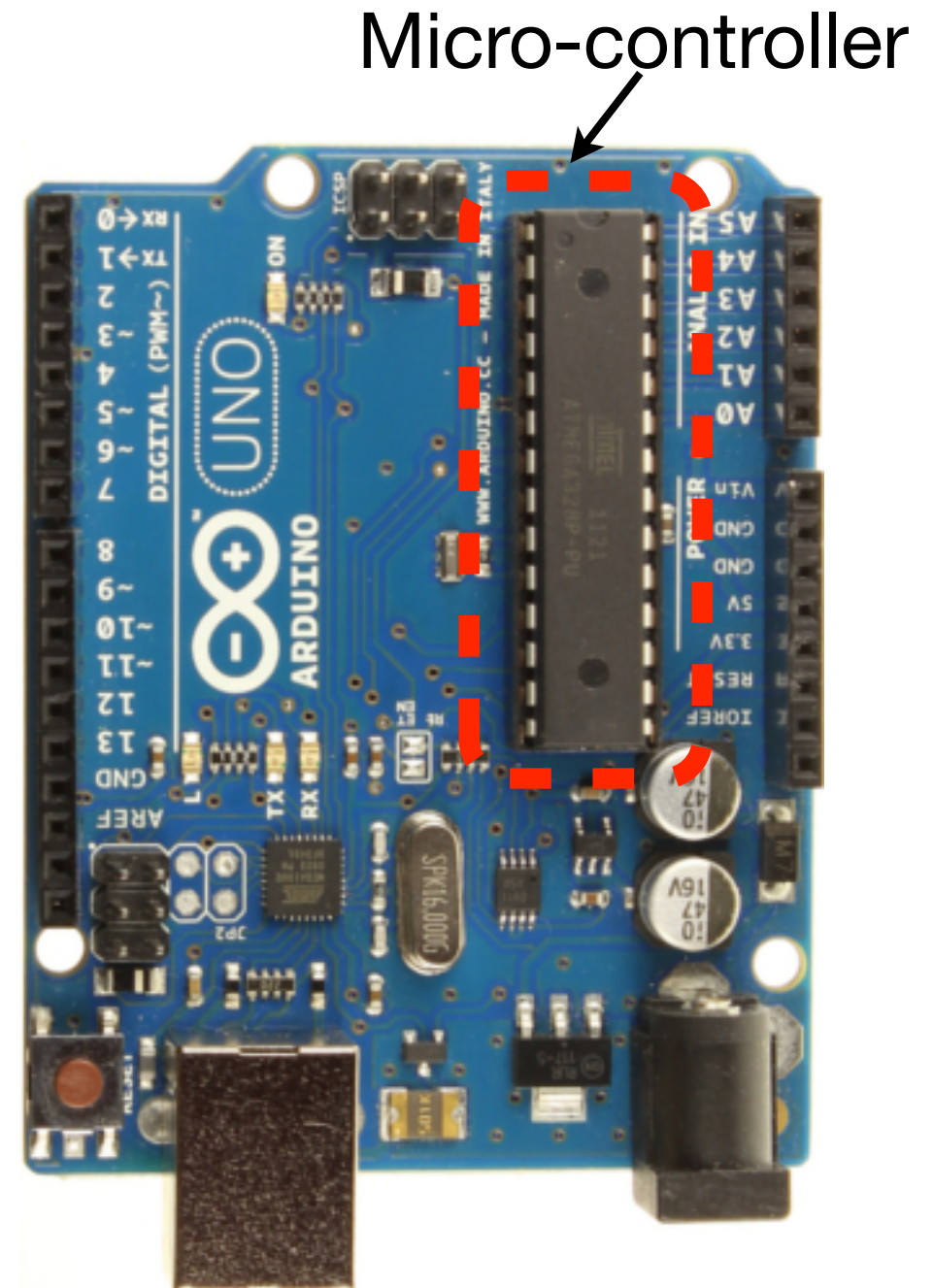
How? With a micro-controller...

- It's just a tiny computer.
- On your computer you program something (in the C language)
- You put that program into the board
- The micro-controller will run that program as long as it has power.
- Your program can:
 - ⦿ Turn ON/OFF pins.
 - ⦿ Read ON/OFF data from pins.

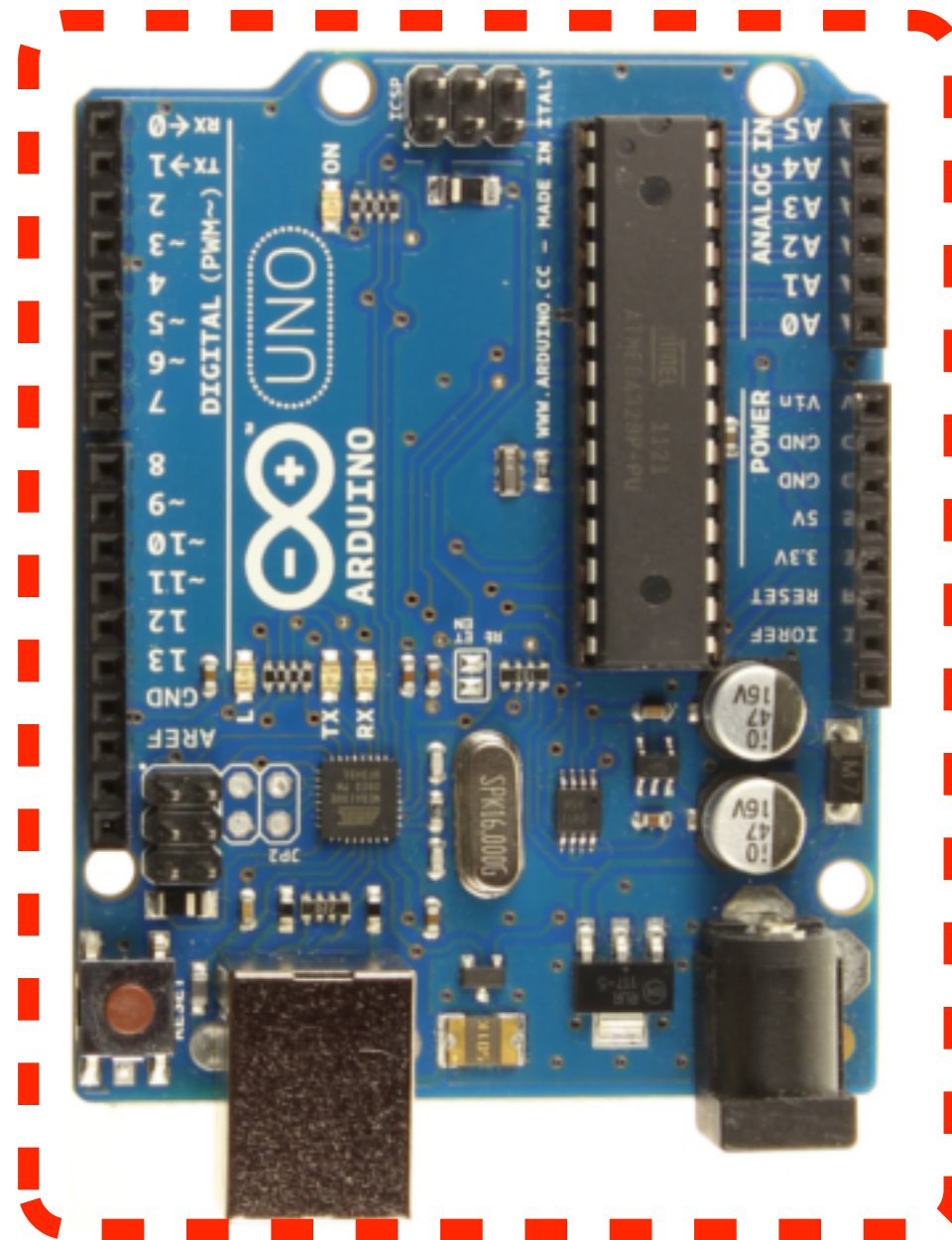


Micro-controller?

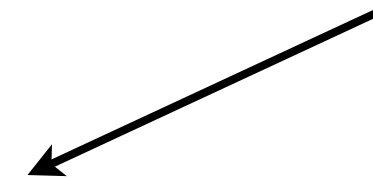
- Pretty slow (16MHz)
- Little memory.
- No operating system.
- No way you can put windows 7 in there.
- Micro-controllers are all over the place: cars, micro-waves, projectors, remote controls...
- ❖ Look around... Where can you find micro-controllers in this room?



We will use the Arduino in our camp!

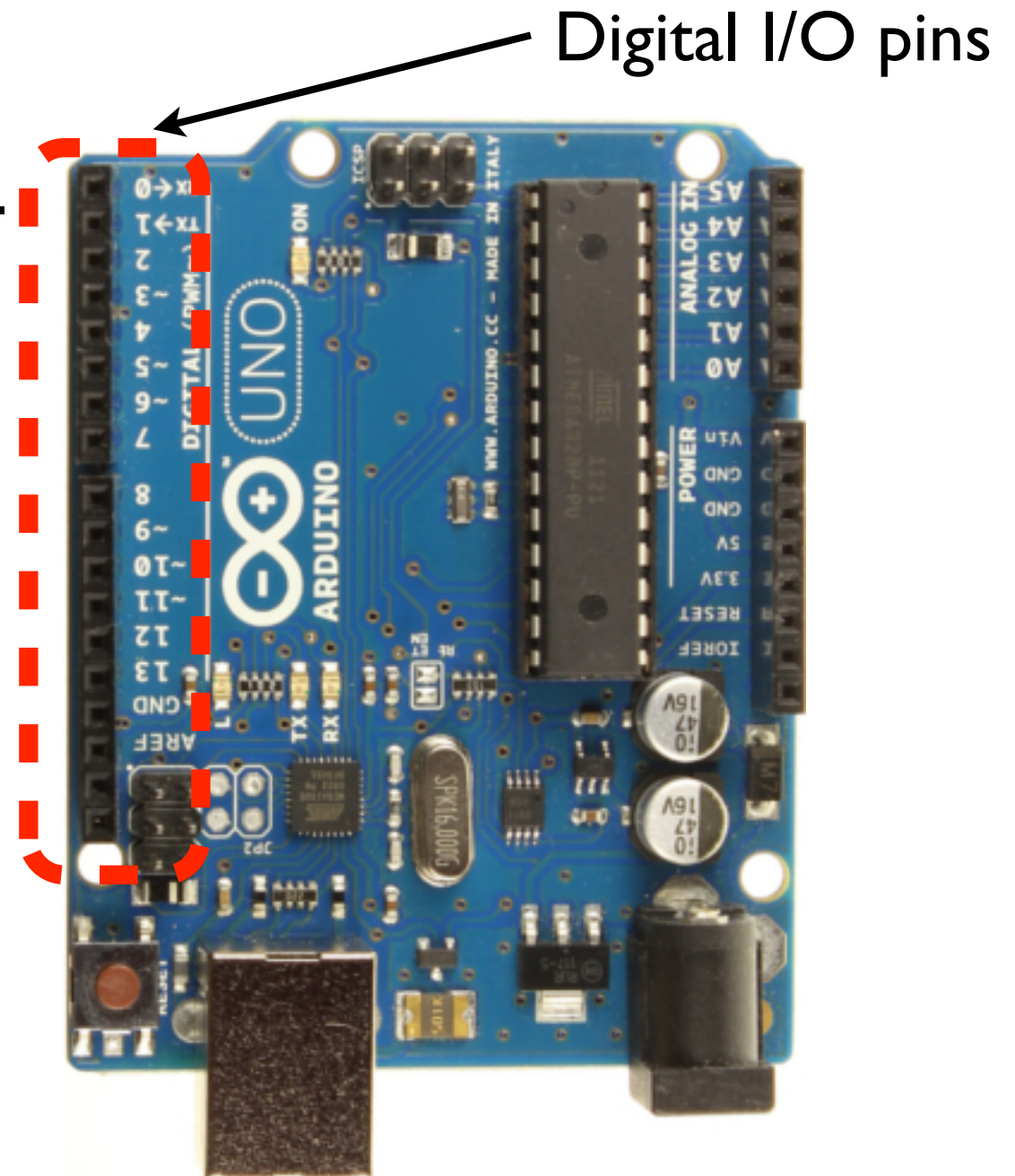


This board is called
Arduino



Digital I/O with the Arduino

- Input/Output (I/O) is done through pins.
- Input = read stuff from the real-world.
- Output = write stuff to the real-world.
- You put a wire into a pin and connect it to something else.
- You say if a pin is **input** or **output**.
- A Digital pins can have two values: **HIGH** or **LOW**.





Put your hacker hat on!

- This means you are about to make something...
- Hackers help each others... Not going forward till everyone is done!
- A hacker is a person who modifies stuff.
- Building, rebuilding, modifying, and creating software, hardware.
- Other kids wish things... YOU make them!
- “I wish I could control my PS3 with my brain.” Well, YOU can do it!





Install the Arduino software

- <http://arduino.cc>
- <http://arduino.cc/it/Guide/Windows>

Yeah...read these instructions!

```
sketch_jan27a | Arduino 1.0

sketch_jan27a $
void setup()
{
  //start serial port at 9600 bps:
  Serial.begin(9600);
}

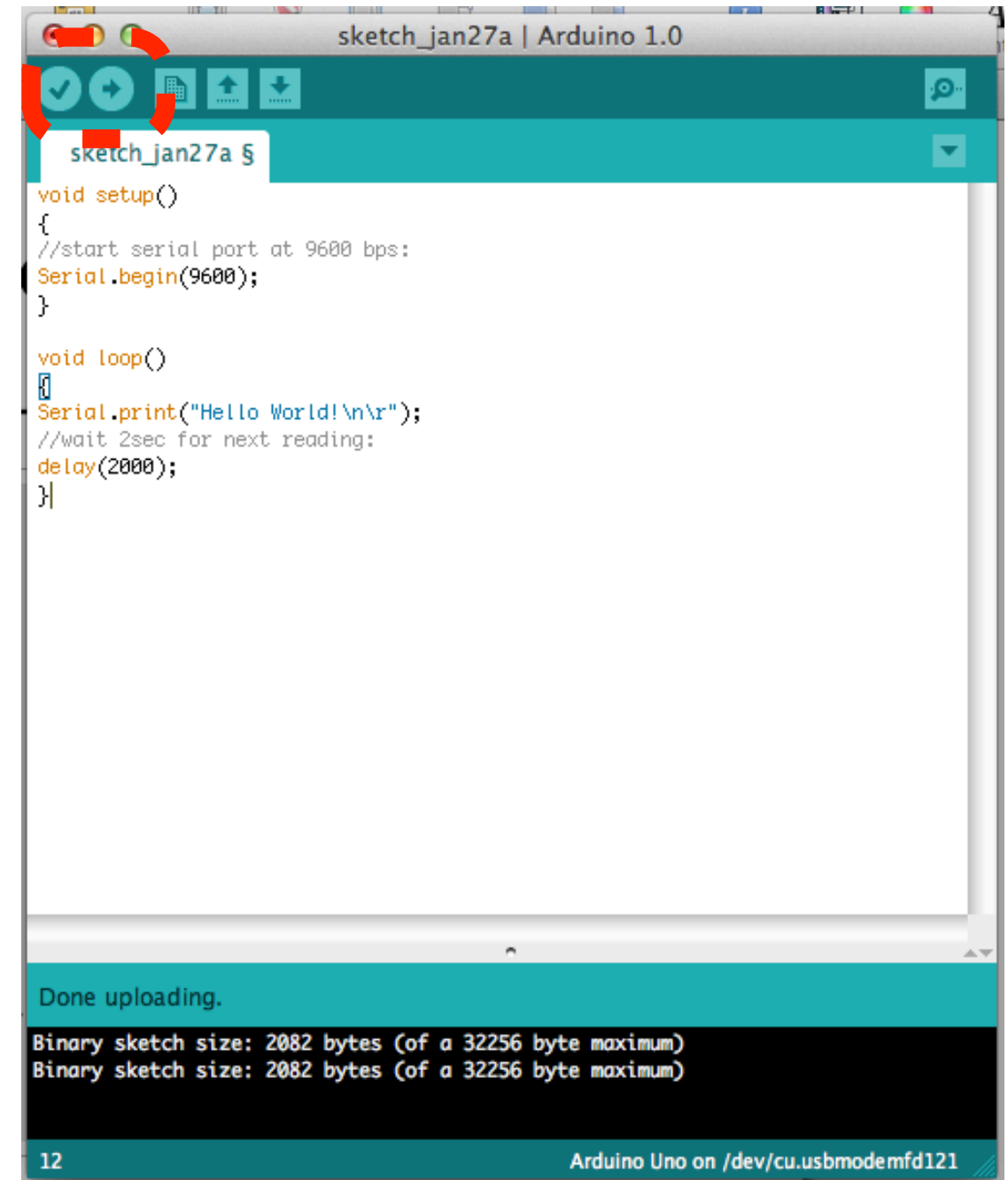
void loop()
{
  Serial.print("Hello World!\n\r");
  //wait 2sec for next reading:
  delay(2000);
}
```

Done uploading.
Binary sketch size: 2082 bytes (of a 32256 byte maximum)
Binary sketch size: 2082 bytes (of a 32256 byte maximum)

12 Arduino Uno on /dev/cu.usbmodemfd121

Arduino software

- You connect the Arduino into your computer with the USB cable.
- You write some stuff here.
- You press the  arrow to send your stuff to the Arduino.

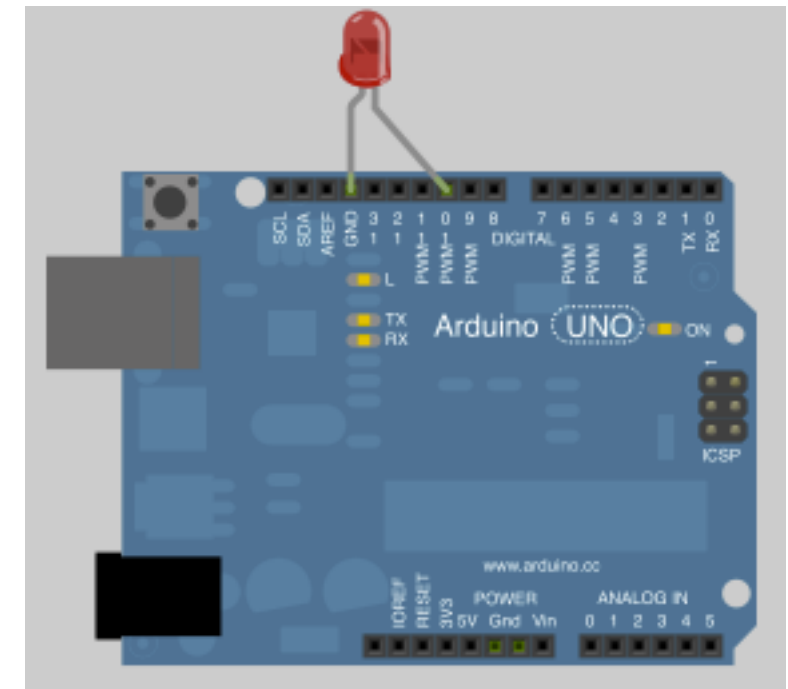




Turn ON/OFF an LED

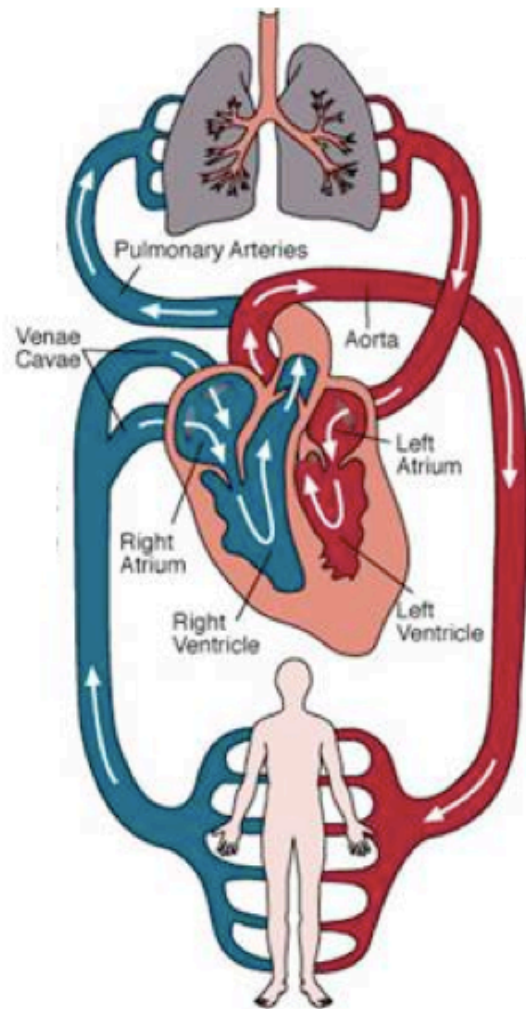
```
int ledPin = 10;
void setup()
{
    pinMode(ledPin, OUTPUT);
}

void loop()
{
    digitalWrite(ledPin, HIGH);
    delay(1000);
    digitalWrite(ledPin, LOW);
    delay(1000);
}
```

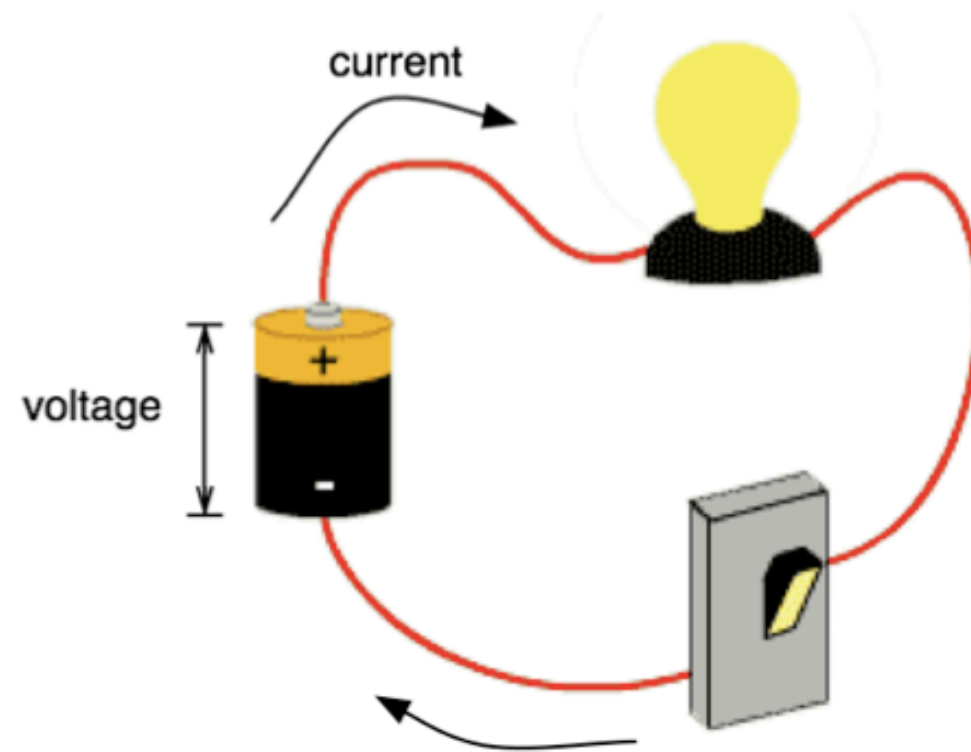


- Connect a LED to pin 10 and ground!
- Type the code.
- Make it blink faster!

What is an electrical circuit



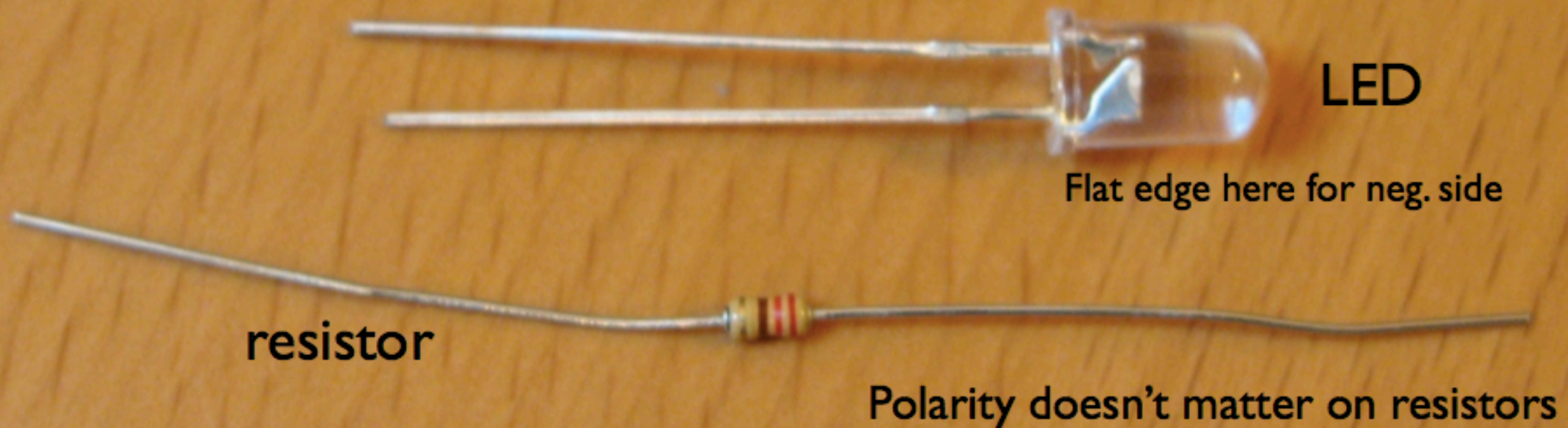
Heart pumps,
Blood flows



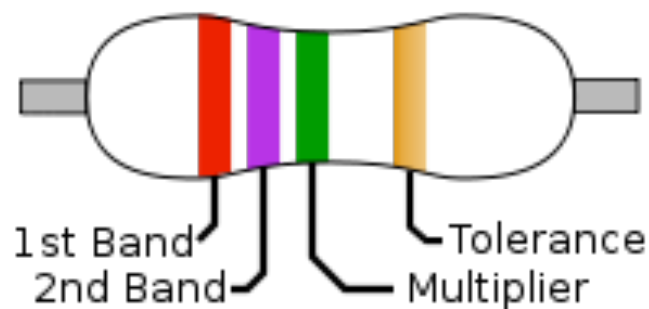
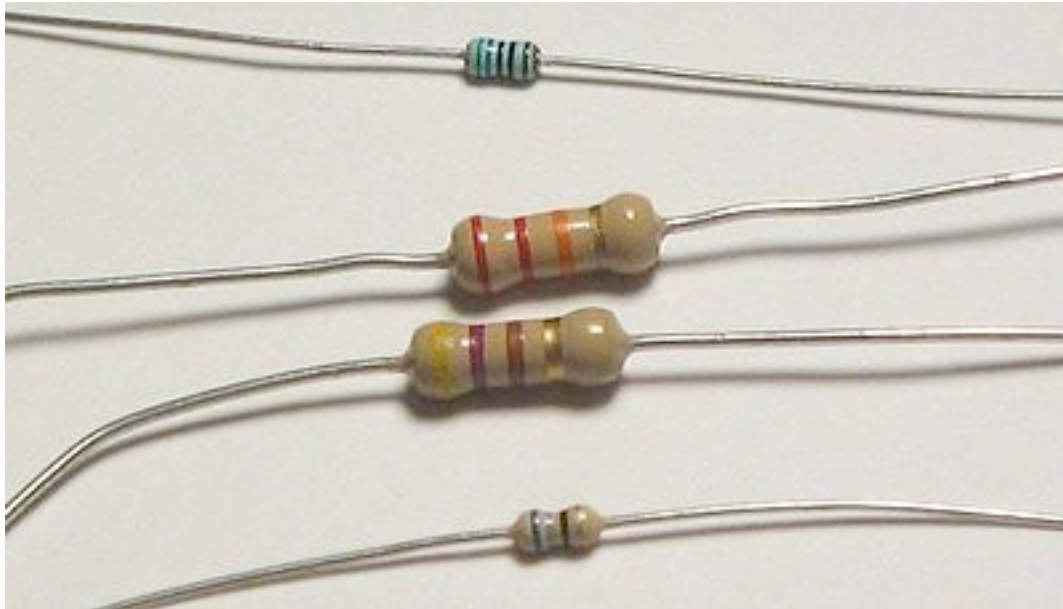
Voltage pushes,
Current flows

LEDs and resistors

On LEDs, polarity matters.
Shorter lead is “negative” side, goes to ground



Resistors?

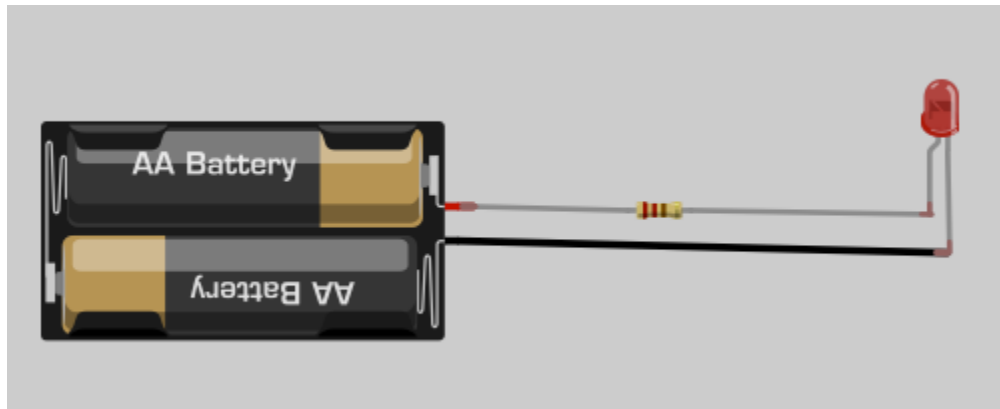


- Think of current as cars.
- Resistors are then bumps in the road.
- Big resistance, means lots of bumps.
- Lots of bumps, the cars (current) will slow down...
- ❖ Why do we need to decrease current?



Lighting an LED

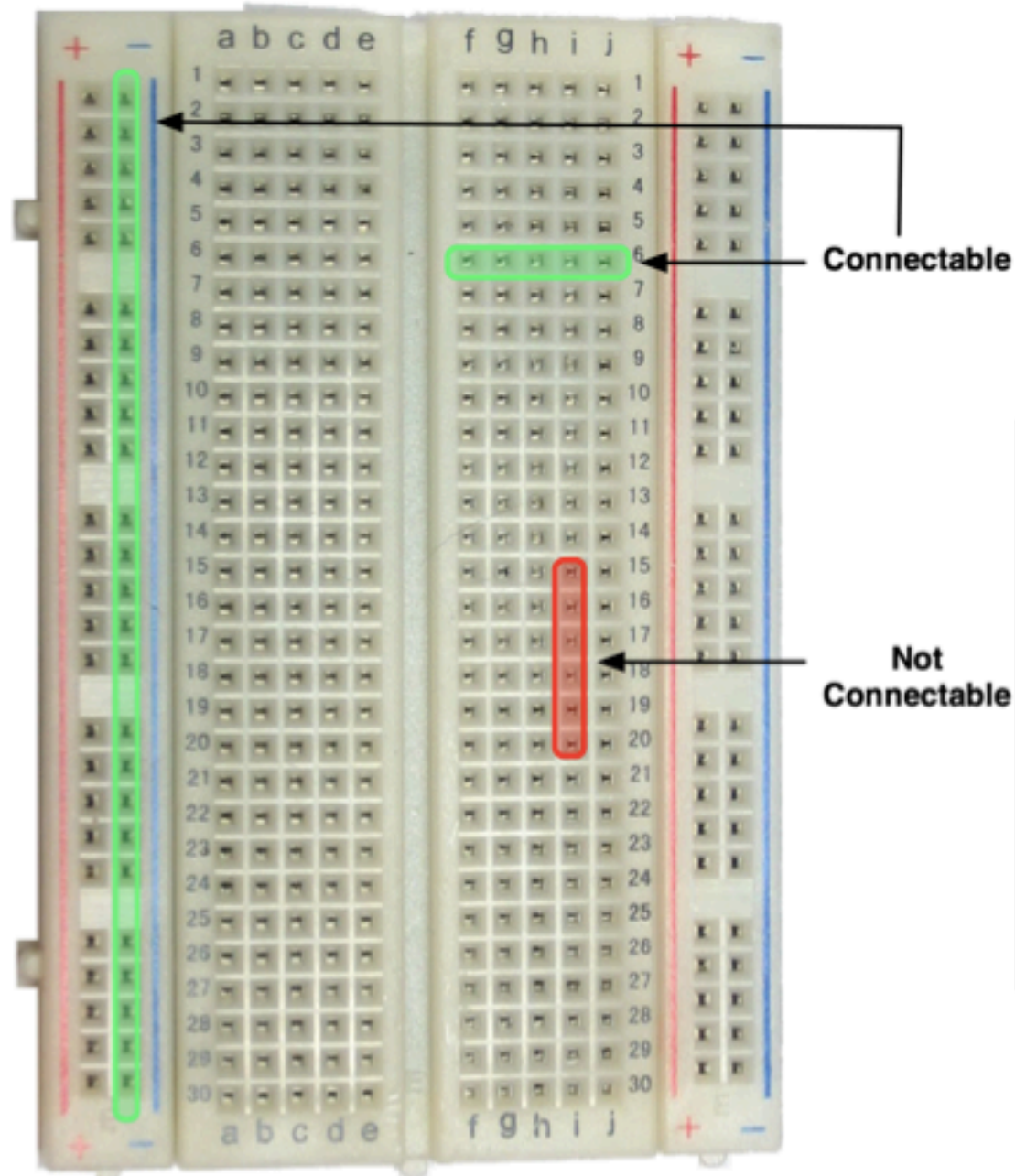
- A hacker **ALWAYS** questions everyone until he tries by himself.
- Too much current destroys.
- So... resistors protect the circuit by reducing the current.



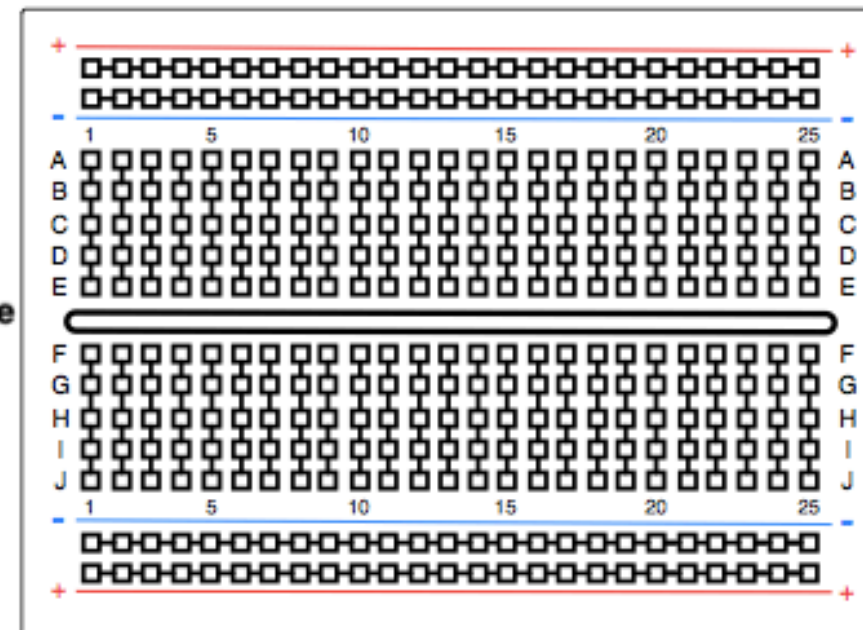
- Attach the LED into a battery without a resistor.
- Keep changing the resistors.
- Use a multi-meter to measure resistance.



We use solderless breadboards to hold circuits in place!

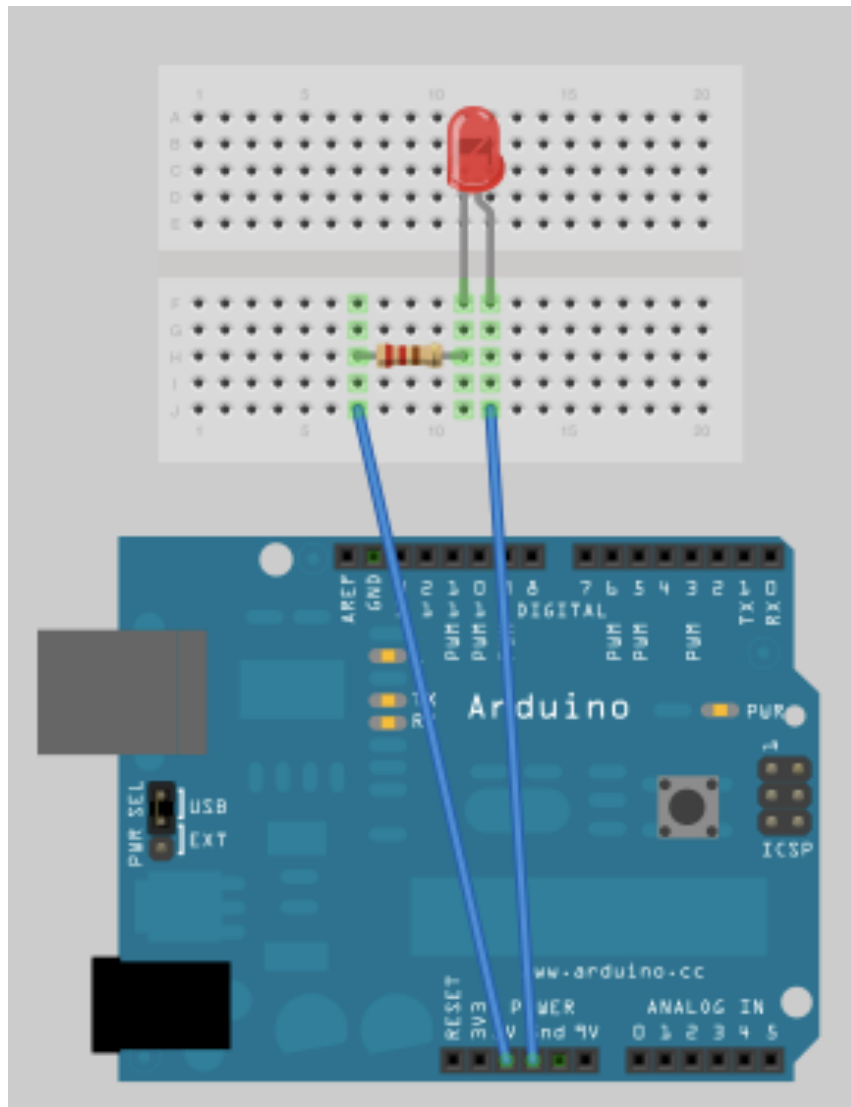


Solderless Breadboard

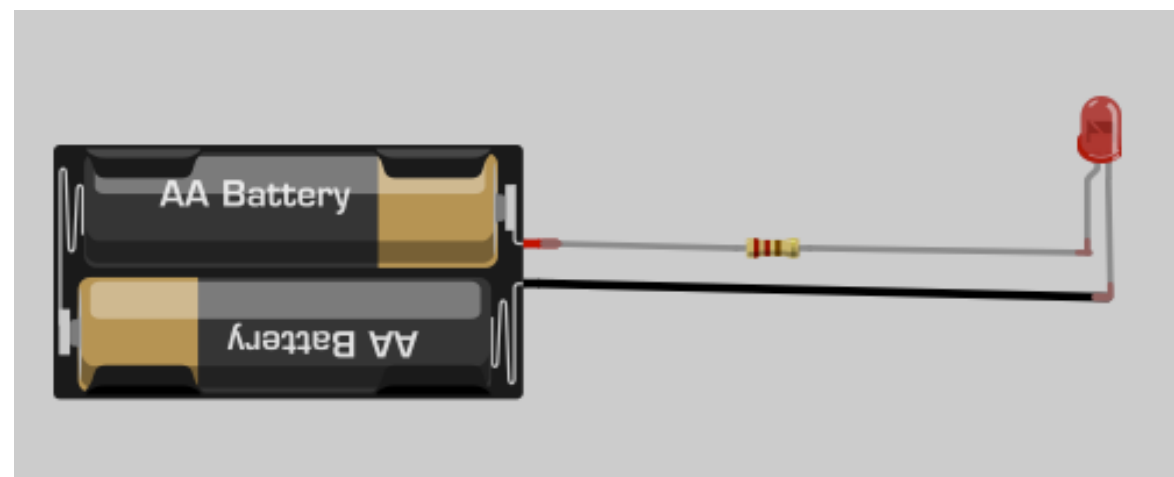




Playing with breadboards

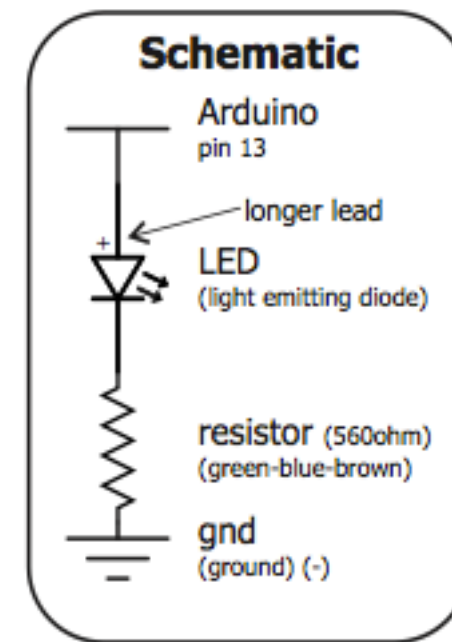
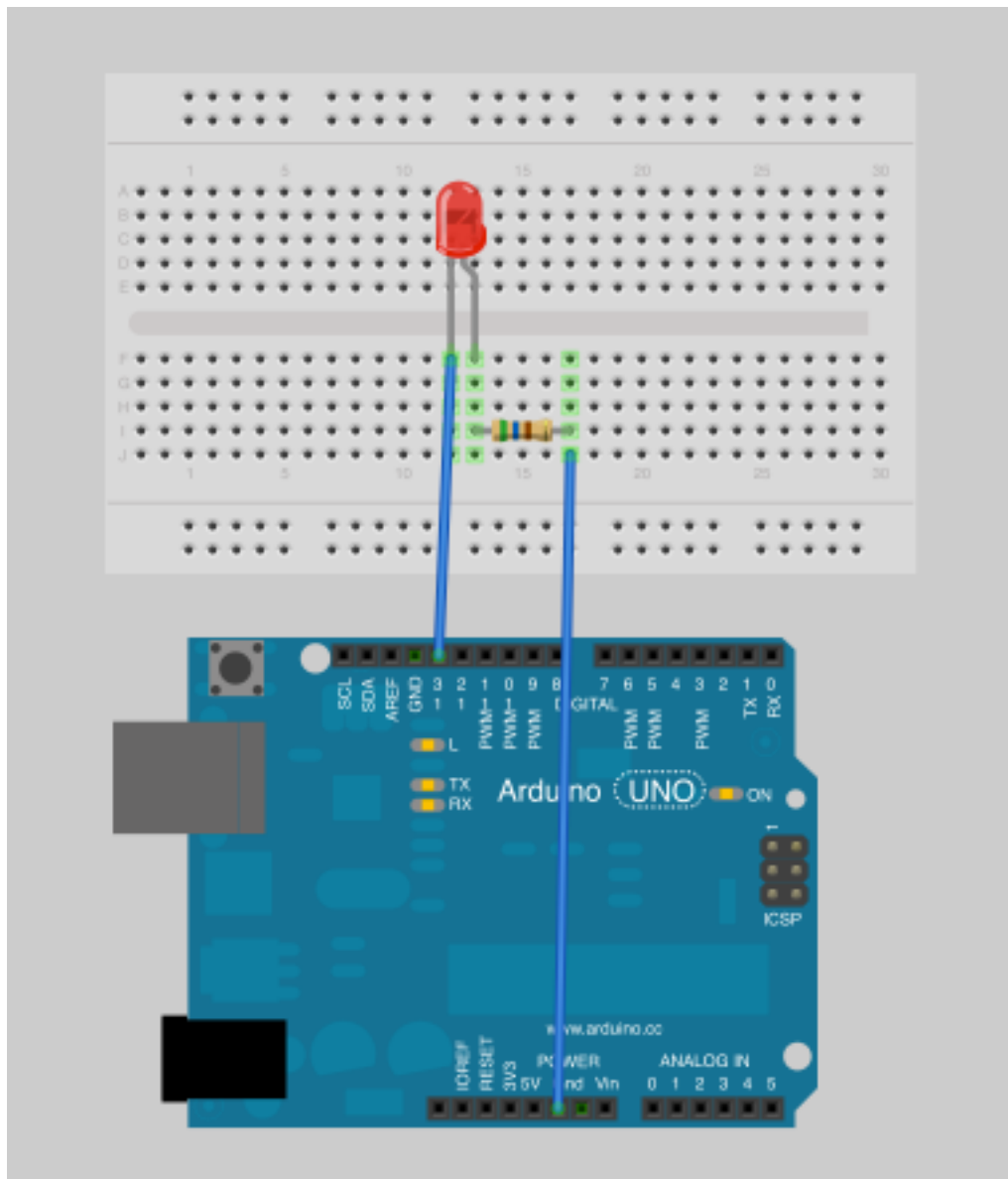


- Connect an LED and a resistor between 5V and GND.
- Pick a resistor that makes the LED really dim and blink it every 2 seconds on pin #3!



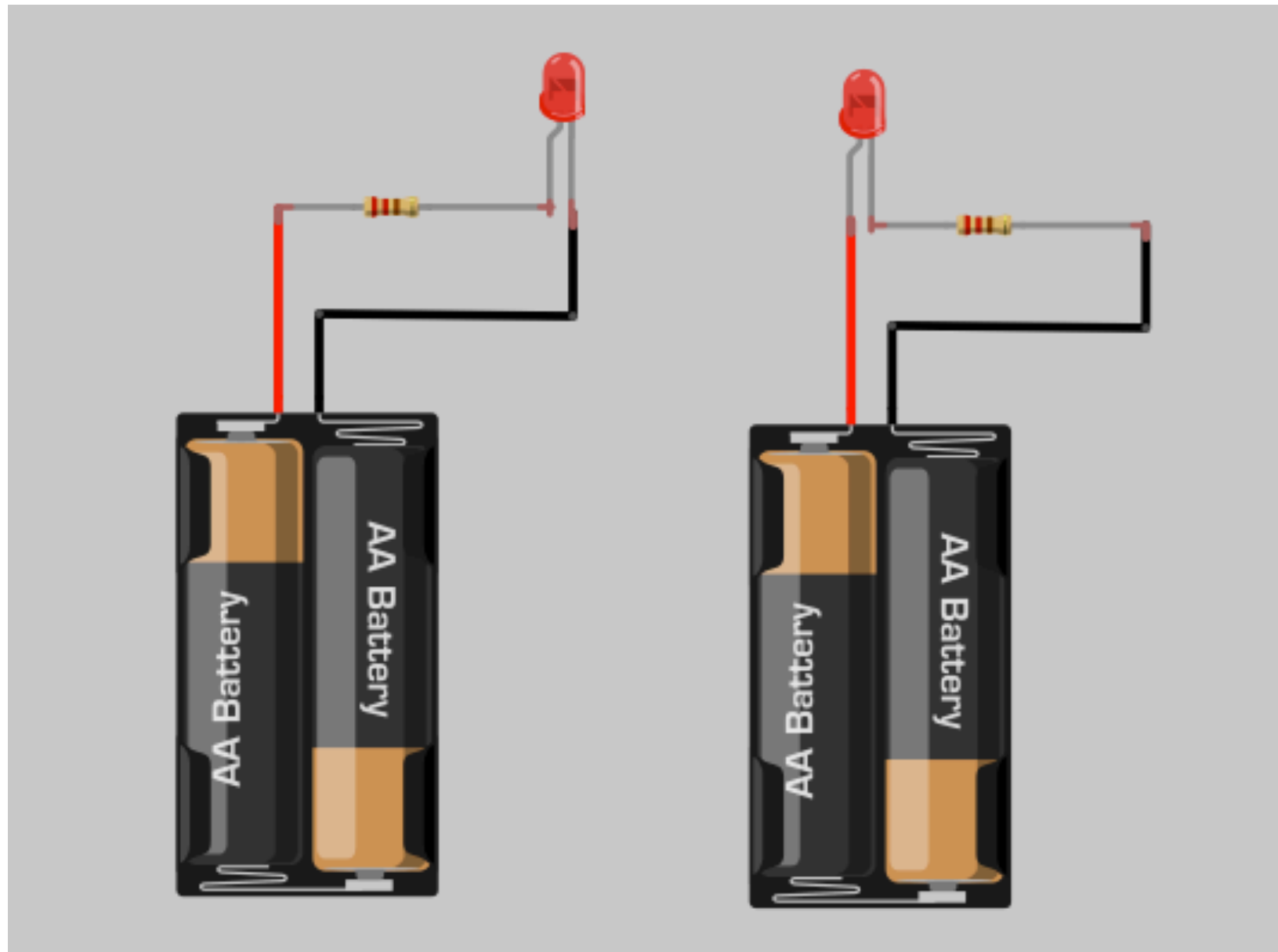


Always add resistors

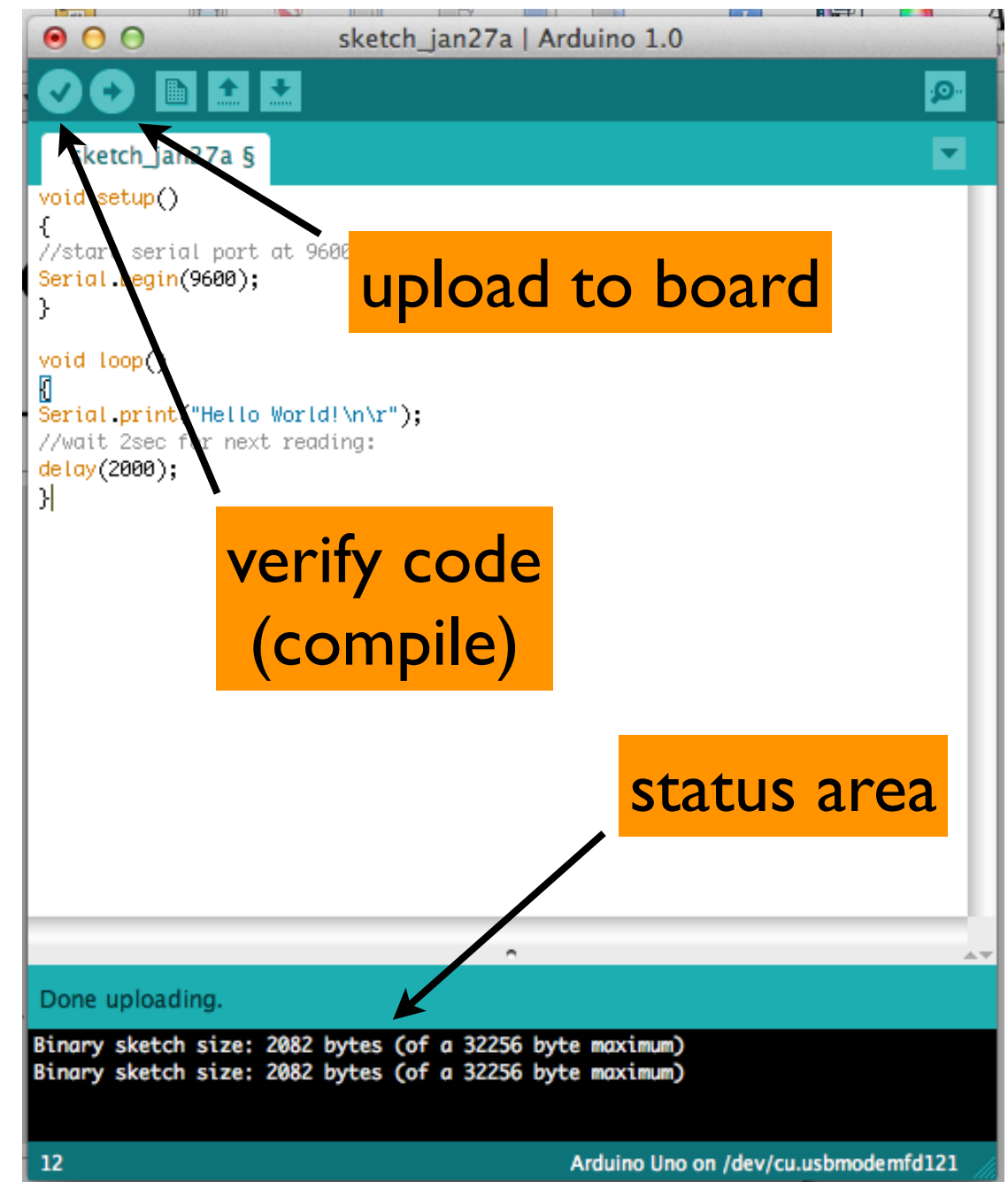
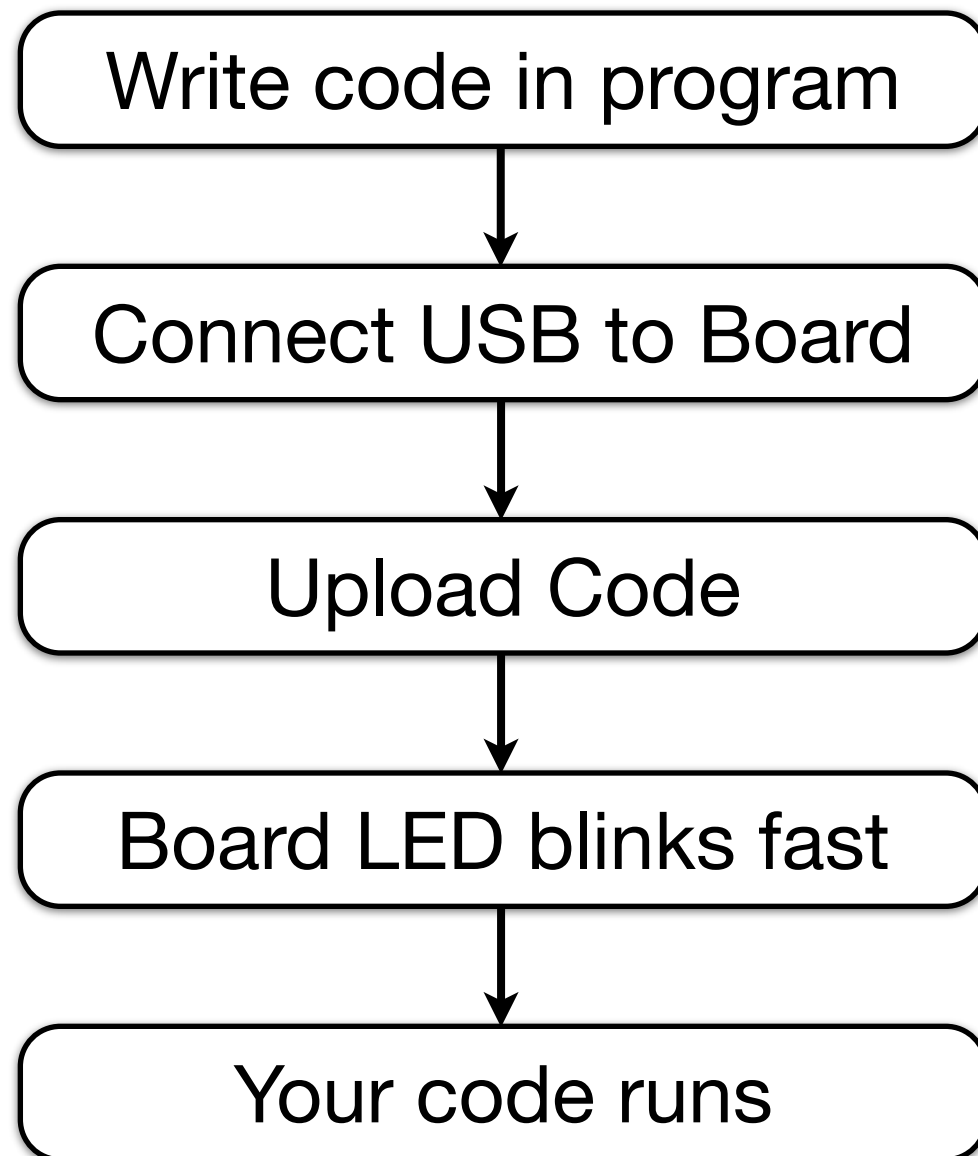


- The schematic and wiring diagram show the same thing.
- Write the code that will blink the LED on pin13 every 5 seconds.

There is no difference between these two circuits



Arduino programming flow



Anatomy of a “sketch”

Global Variables

Declare variables on top...
a variable is like a box of stuff.

Setup()

Stuff that is executed only once at
beginning (eg: set pins)

Loop()

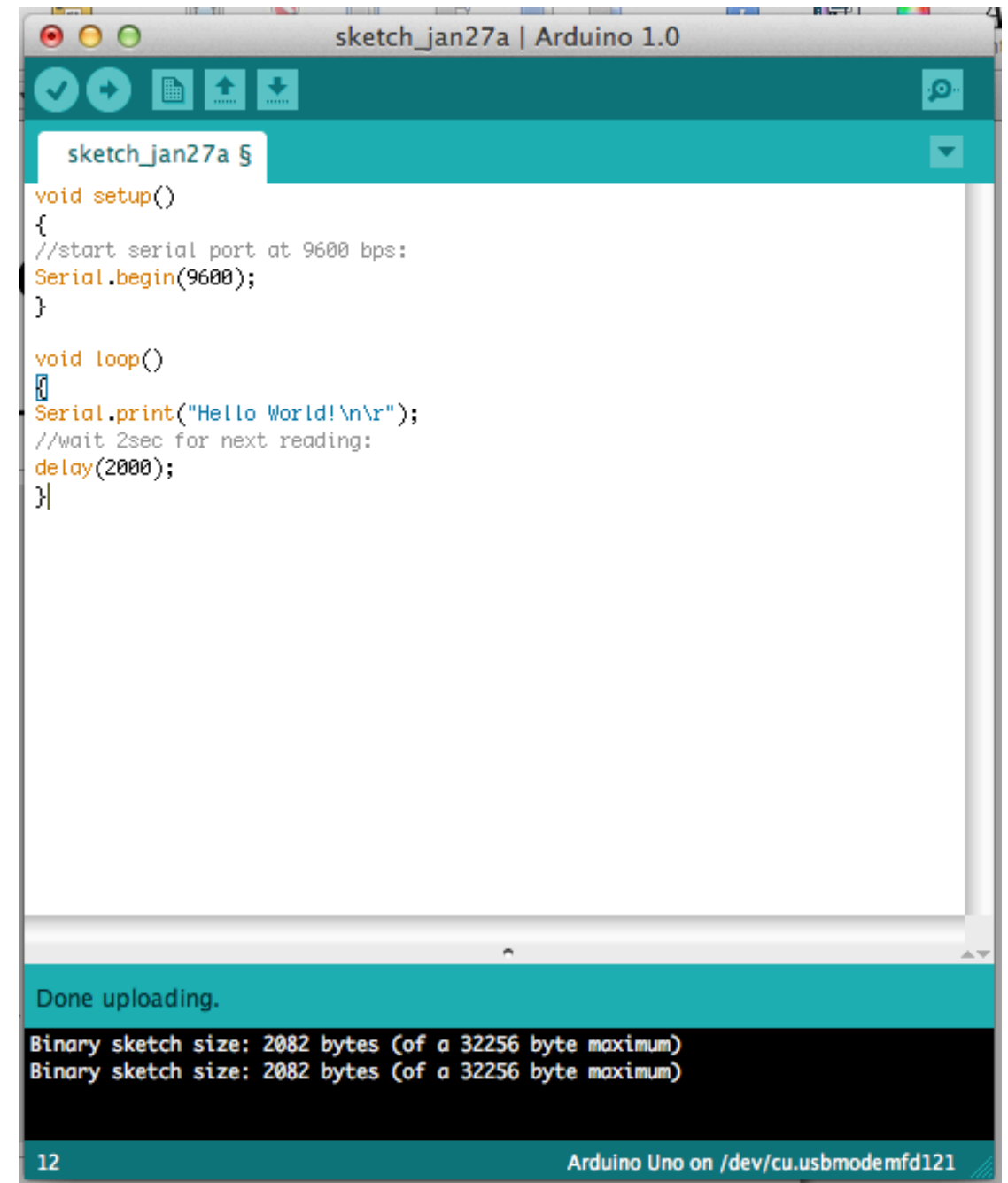
Stuff that runs repeatedly, after
setup()

Arduino language

You can tell the Arduino to do lots of useful stuff.

This is done with functions:

- *pinMode()* – set a pin as input or output
- *digitalWrite()* – set a digital pin high/low
- *digitalRead()* – read a digital pin's state
- *delay()* – wait an amount of time
- *millis()* – get the current time



```
sketch_jan27a | Arduino 1.0

sketch_jan27a $

void setup()
{
  //start serial port at 9600 bps:
  Serial.begin(9600);
}

void loop()
{
  Serial.print("Hello World!\n\r");
  //wait 2sec for next reading:
  delay(2000);
}
```

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12 Arduino Uno on /dev/cu.usbmodemfd121

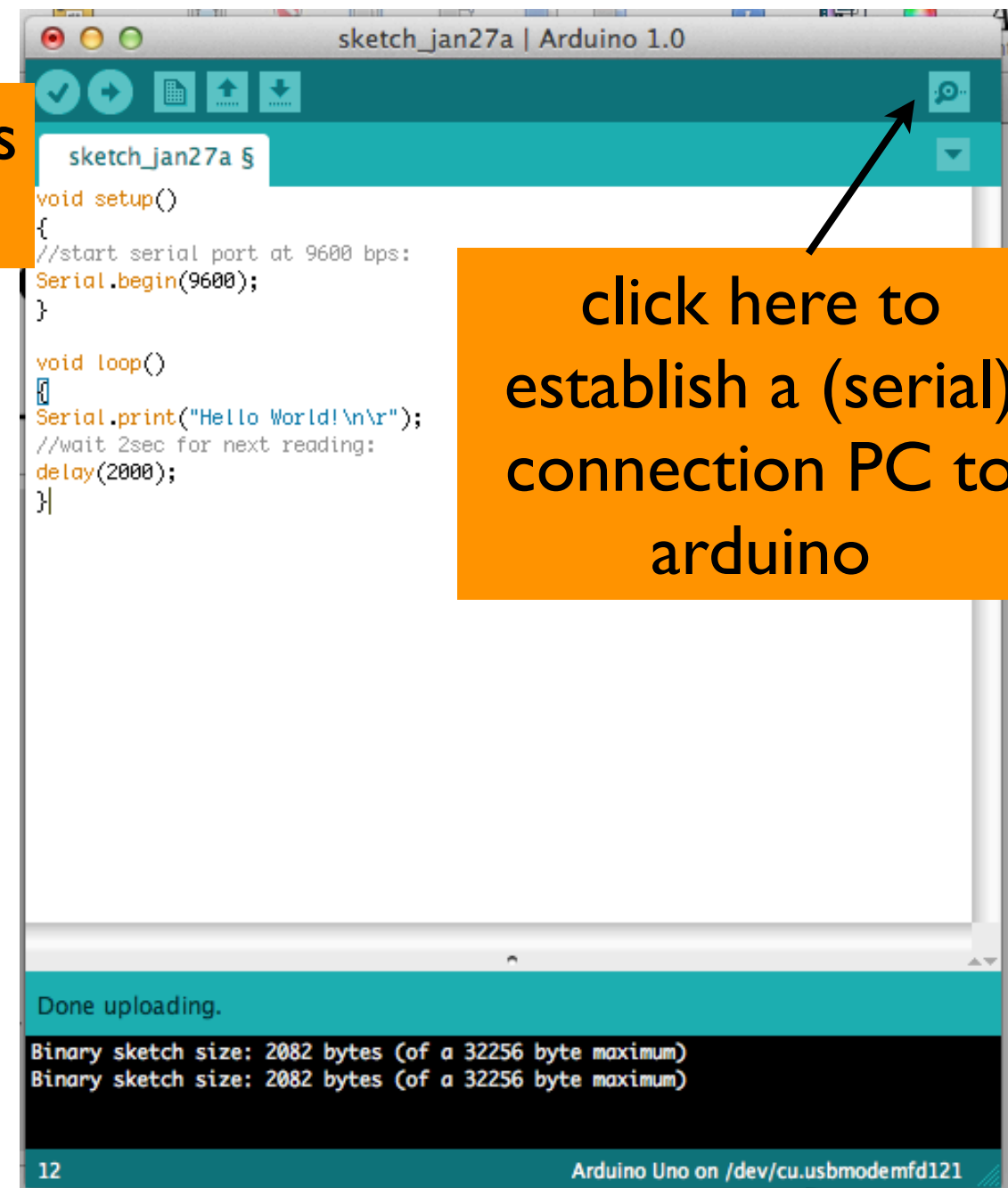


“Hello World”

```
void setup()
{
  //start serial port at 9600 bps
  Serial.begin(9600);
}

void loop()
{
  //print something on the computer
  Serial.print("Hello World!\n");
  //wait 2sec
  delay(2000);
}
```

the Arduino ignores
anything after //



click here to
establish a (serial)
connection PC to
arduino

ASCII table and serial connections

Dec	Hex	Char	Dec	Hex	Char	Dec	Hex	Char	Dec	Hex	Char
0	00	Null	32	20	Space	64	40	@	96	60	`
1	01	Start of heading	33	21	!	65	41	A	97	61	a
2	02	Start of text	34	22	"	66	42	B	98	62	b
3	03	End of text	35	23	#	67	43	C	99	63	c
4	04	End of transmit	36	24	\$	68	44	D	100	64	d
5	05	Enquiry	37	25	%	69	45	E	101	65	e
6	06	Acknowledge	38	26	&	70	46	F	102	66	f
7	07	Audible bell	39	27	'	71	47	G	103	67	g
8	08	Backspace	40	28	(	72	48	H	104	68	h
9	09	Horizontal tab	41	29	)	73	49	I	105	69	i
10	0A	Line feed	42	2A	*	74	4A	J	106	6A	j
11	0B	Vertical tab	43	2B	+	75	4B	K	107	6B	k
12	0C	Form feed	44	2C	,	76	4C	L	108	6C	l
13	0D	Carriage return	45	2D	-	77	4D	M	109	6D	m
14	0E	Shift out	46	2E	.	78	4E	N	110	6E	n
15	0F	Shift in	47	2F	/	79	4F	O	111	6F	o
16	10	Data link escape	48	30	0	80	50	P	112	70	p
17	11	Device control 1	49	31	1	81	51	Q	113	71	q
18	12	Device control 2	50	32	2	82	52	R	114	72	r
19	13	Device control 3	51	33	3	83	53	S	115	73	s
20	14	Device control 4	52	34	4	84	54	T	116	74	t
21	15	Neg. acknowledge	53	35	5	85	55	U	117	75	u
22	16	Synchronous idle	54	36	6	86	56	V	118	76	v
23	17	End trans. block	55	37	7	87	57	W	119	77	w
24	18	Cancel	56	38	8	88	58	X	120	78	x
25	19	End of medium	57	39	9	89	59	Y	121	79	y
26	1A	Substitution	58	3A	:	90	5A	Z	122	7A	z
27	1B	Escape	59	3B	;	91	5B	[	123	7B	{
28	1C	File separator	60	3C	<	92	5C	\	124	7C	
29	1D	Group separator	61	3D	=	93	5D	]	125	7D	}
30	1E	Record separator	62	3E	>	94	5E	^	126	7E	~
31	1F	Unit separator	63	3F	?	95	5F	_	127	7F	□

- It is possible to control the Arduino through a serial connection
- Every time you press a key, the computer converts it into a number
- Each key has a different number and this is called ASCII.
- We can send that number to the Arduino through the cable.



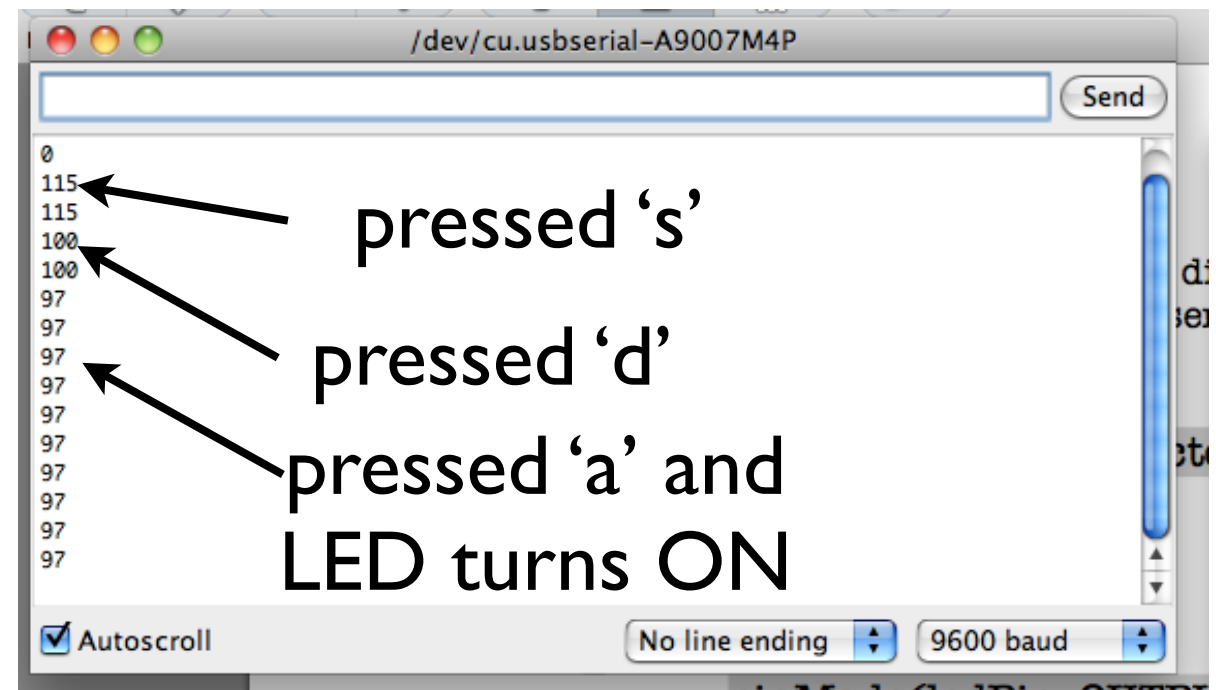
Character 'a' turns on LED

```
int ledPin = 10; int inByte = 0;
void setup()
{
  pinMode(ledPin, OUTPUT);
  Serial.begin(9600);
}

void loop()
{
  while (Serial.available() > 0)
  { inByte = Serial.read(); }

  Serial.println(inByte);

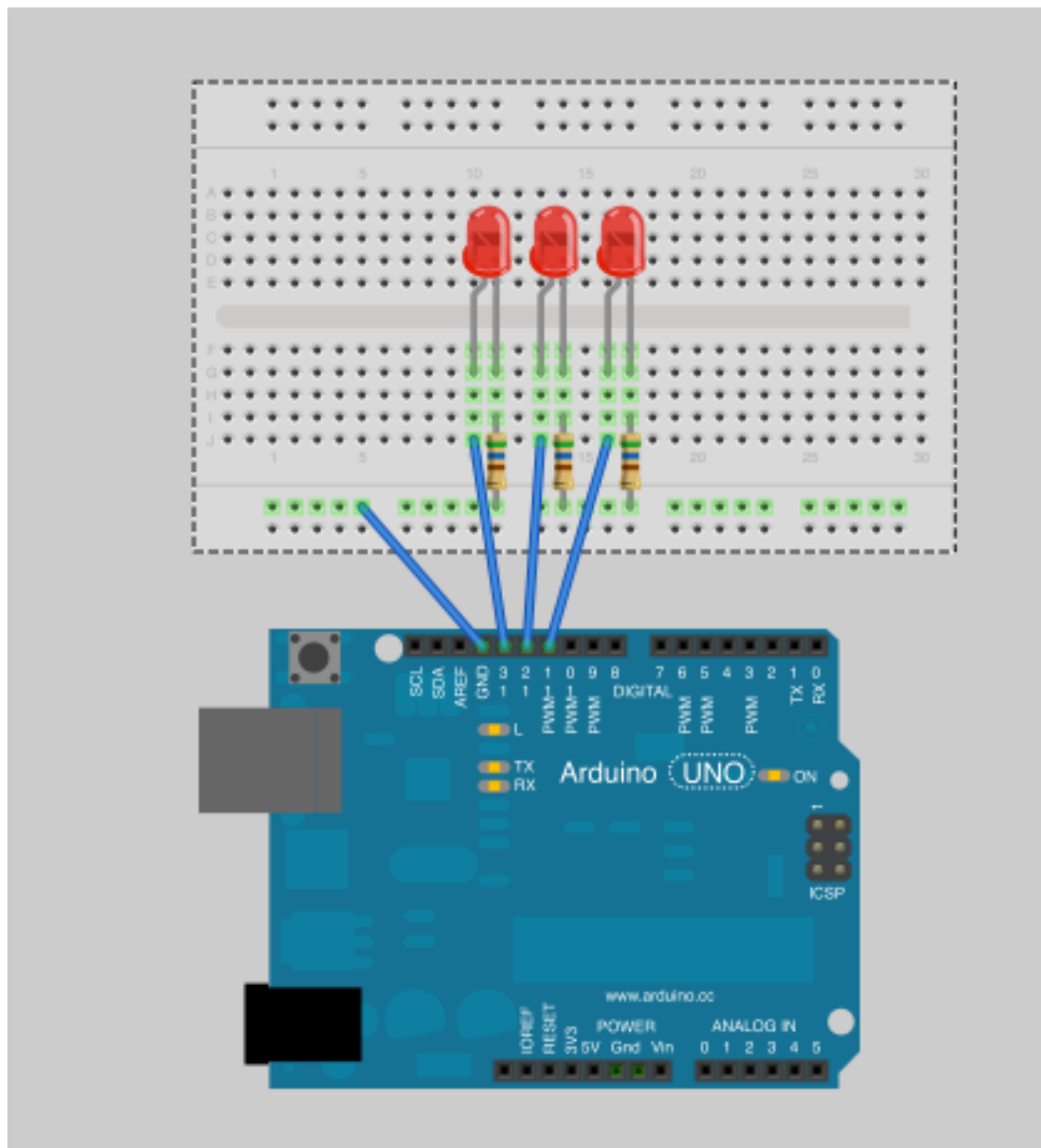
  if (inByte == 97)
  { digitalWrite(ledPin, HIGH); }
  else
  { digitalWrite(ledPin, LOW); }
  delay(1000);
}
```



- Attach a LED to pin 10 and run this code.
- Now attach another LED to pin 7 and modify the code.
- Whenever 'b' is pressed, this new LED should go ON.



Dealing with 3 LEDs



- Put this circuit together

When:

- 'a' is pressed, only LED#1 is ON
- 'b' is pressed, only LED#2 is ON
- 'c' is pressed, only LED #3 is ON

Dealing with 3 LEDs (possible code)

```
int ledPinA = 11, ledPinB = 12,  
ledPinC = 13;  
int inByte = 0;
```

```
void setup()  
{  
  pinMode(ledPinA, OUTPUT);  
  pinMode(ledPinB, OUTPUT);  
  pinMode(ledPinC, OUTPUT);  
  Serial.begin(9600);  
}
```

```
void loop(){  
  while (Serial.available()>0)  
  { inByte = Serial.read(); }  
  Serial.println(inByte);  
  
  if (inByte==97) {  
    digitalWrite(ledPinA, HIGH);  
    digitalWrite(ledPinB, LOW);  
    digitalWrite(ledPinC, LOW);}  
  if (inByte==98) {  
    digitalWrite(ledPinA, LOW);  
    digitalWrite(ledPinB, HIGH);  
    digitalWrite(ledPinC, LOW);}  
  if (inByte==99) {  
    digitalWrite(ledPinA, LOW);  
    digitalWrite(ledPinB, LOW);  
    digitalWrite(ledPinC, HIGH);}  
  
  delay(1000);  
}
```


Random numbers

- The Arduino can give you random numbers.
- However, you must use a fancy function called *random*.
- ◎ What will the Arduino print out?

```
void setup()
{
  Serial.begin(9600);
}

void loop()
{
  int randNumber;
  //print a number from 0 to 299
  randNumber = random(300);
  Serial.println(randNumber);

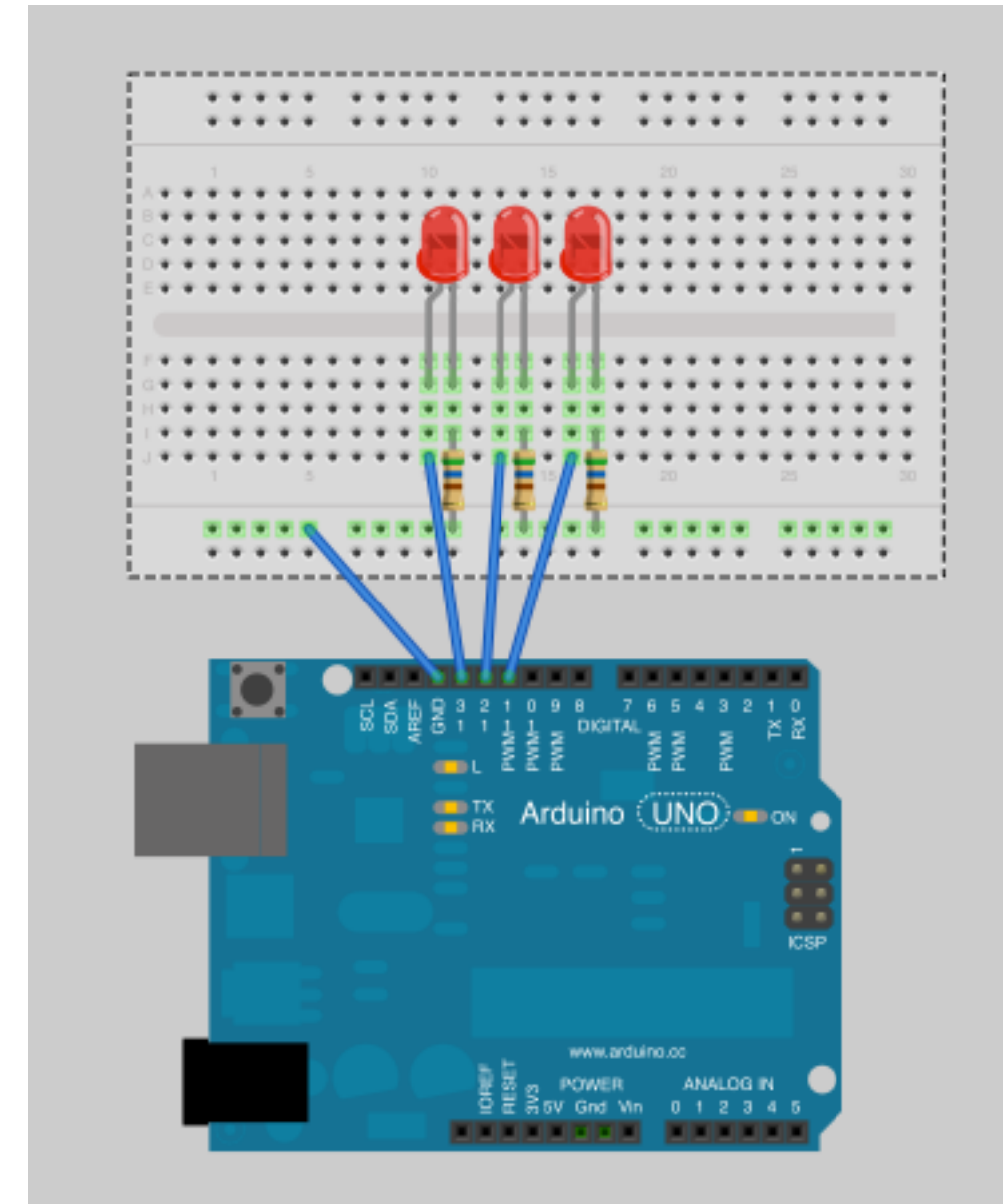
  // print a number from 10 to 19
  randNumber = random(10, 20);
  Serial.println(randNumber);

  delay(1000);
}
```



Circuit with 3 LEDs

- Create a circuit with 3 LEDs in pins 11, 12 and 13.
- If '1' is pressed only one RANDOM LED will be ON.
- If '2' is pressed at most two RANDOM LED's will be ON.
- If '3' is pressed ALL three LED's will be ON.



Circuit with 3 LEDs (possible code)

```
int ledPinA = 11, ledPinB = 12;  
int ledPinC = 13, inByte = 0;
```

```
void setup(){  
  pinMode(ledPinA, OUTPUT);  
  pinMode(ledPinB, OUTPUT);  
  pinMode(ledPinC, OUTPUT);  
  Serial.begin(9600);  
}  
void loop(){  
  while (Serial.available()>0)  
  { inByte = Serial.read(); }  
  Serial.println(inByte);  
  
  if (inByte==49) {  
    digitalWrite(ledPinA, LOW);  
    digitalWrite(ledPinB, LOW);  
    digitalWrite(ledPinC, LOW);  
    int rnd = random(11,14);  
    digitalWrite(rnd, HIGH);  
  }
```

```
    if (inByte==50) {  
      digitalWrite(ledPinA, LOW);  
      digitalWrite(ledPinB, LOW);  
      digitalWrite(ledPinC, LOW);  
      digitalWrite(random(11,14), HIGH);  
      digitalWrite(random(11,14), HIGH);  
    }  
  
    if (inByte==51) {  
      digitalWrite(ledPinA, HIGH);  
      digitalWrite(ledPinB, HIGH);  
      digitalWrite(ledPinC, HIGH);  
    }  
  
    delay(1000);  
  }
```


Cylon

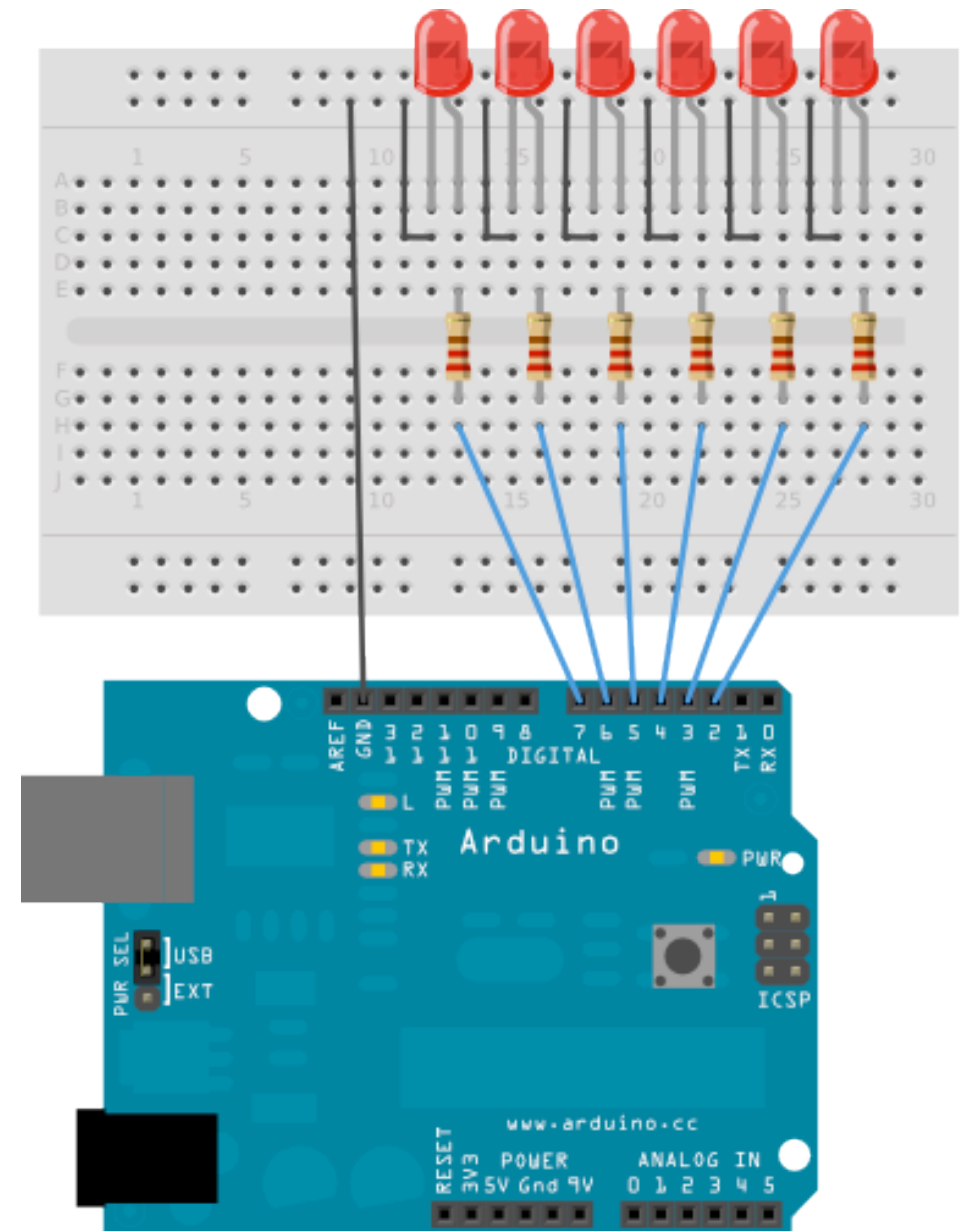


Video - [#1](#) [#2](#)

- BattleStar Gallatica is an awesome TV show.
- A group of robots are trying to destroy all humans.
- Their eyes are LEDS that move in a neat pattern.
- We can do that pattern too!

Cylon eyes

- Often you want to iterate over a series of pins and do something to each one.
- We can use a `for()` loop to cycle back and forth through digital pins 2-7.
- The LEDs are turned on and off, in sequence, by using both the `digitalWrite()` and `delay()` functions.



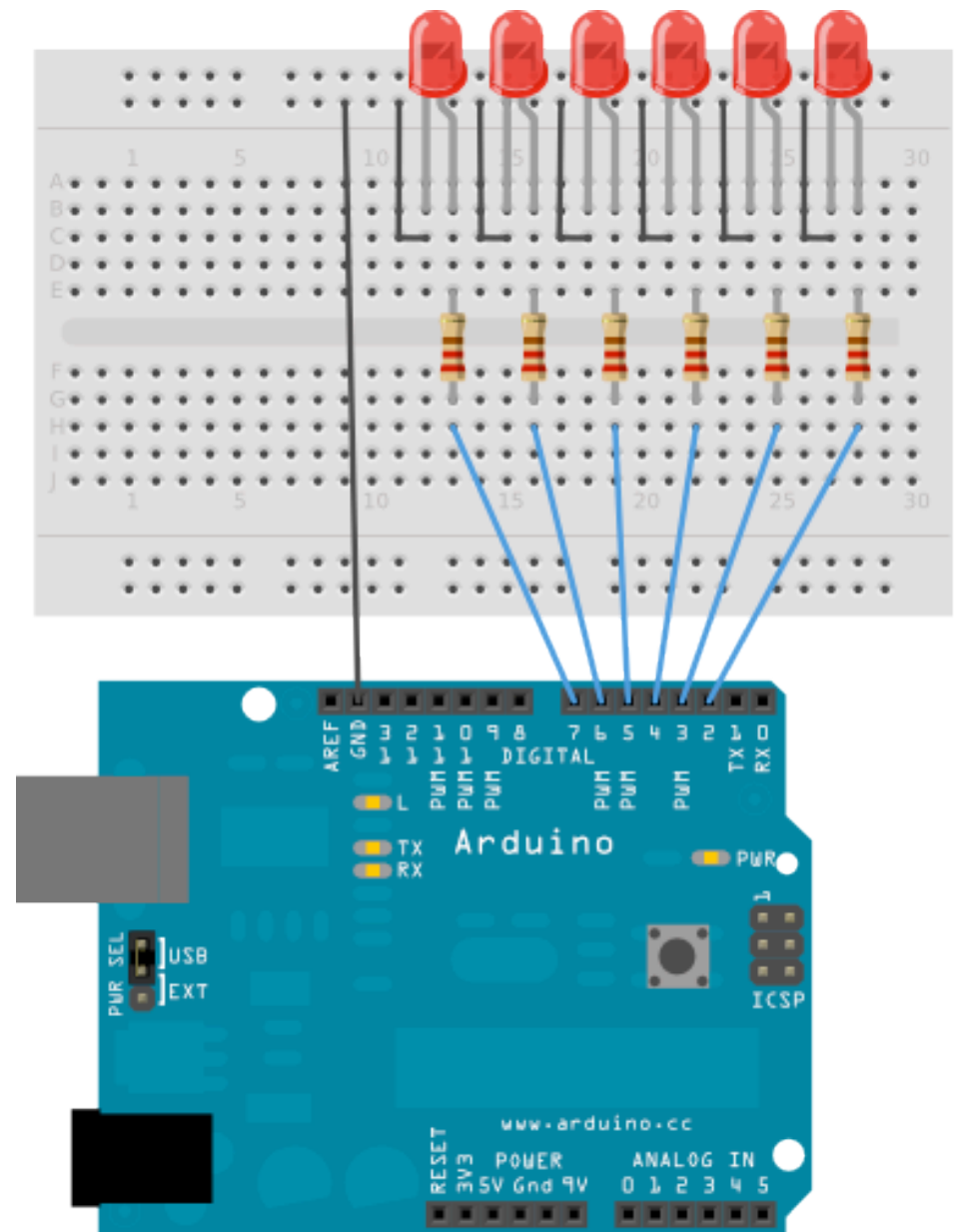


Make your own Cylon eyes

```
//Higher number, means slower timing.
int timer = 100;

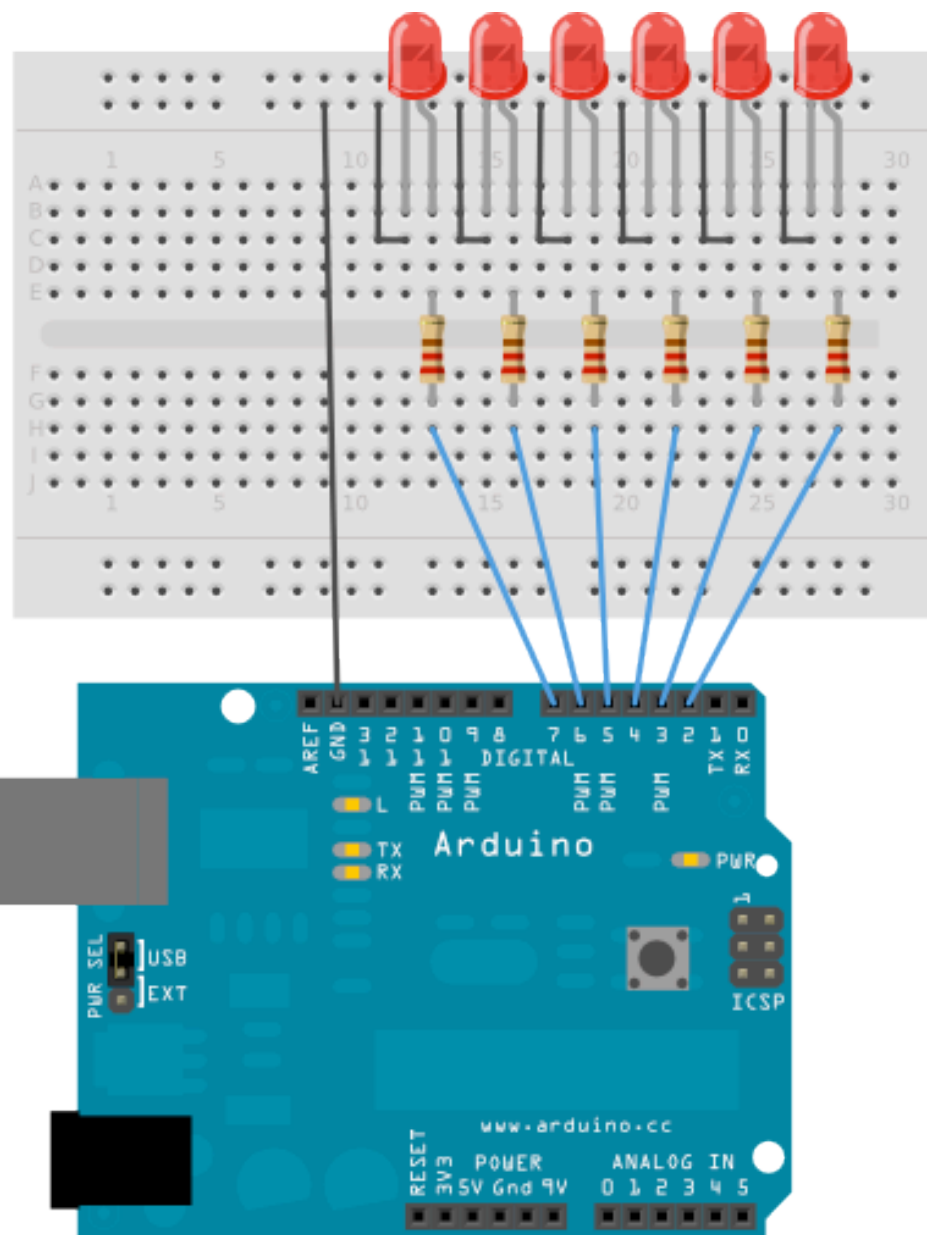
void setup() {
  //for loop to make each pin an output
  for (int thisPin=2; thisPin<8; thisPin++)
  {
    pinMode(thisPin, OUTPUT);
  }
}

void loop() {
  // loop from lowest to highest pin
  for (int thisPin=2; thisPin<8; thisPin++)
  {
    digitalWrite(thisPin, HIGH);
    delay(timer);
    digitalWrite(thisPin, LOW);
  }
}
```





Tweak the Cylon



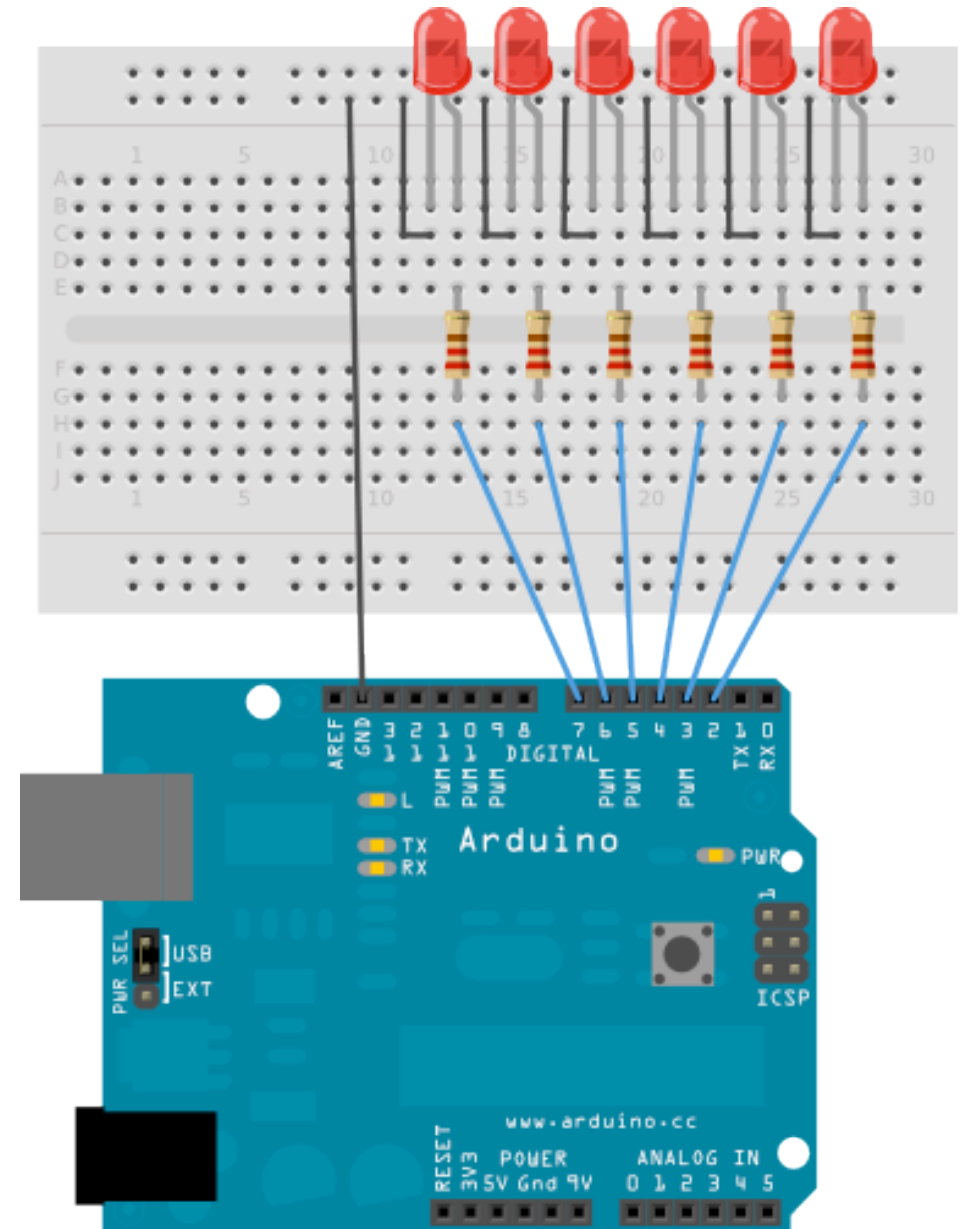
- Make the LEDS cycle LEFT to RIGHT and then RIGHT to LEFT.
- Change the sweeping speed.

Tweak the Cylon (possible code)

```
int timer = 50;

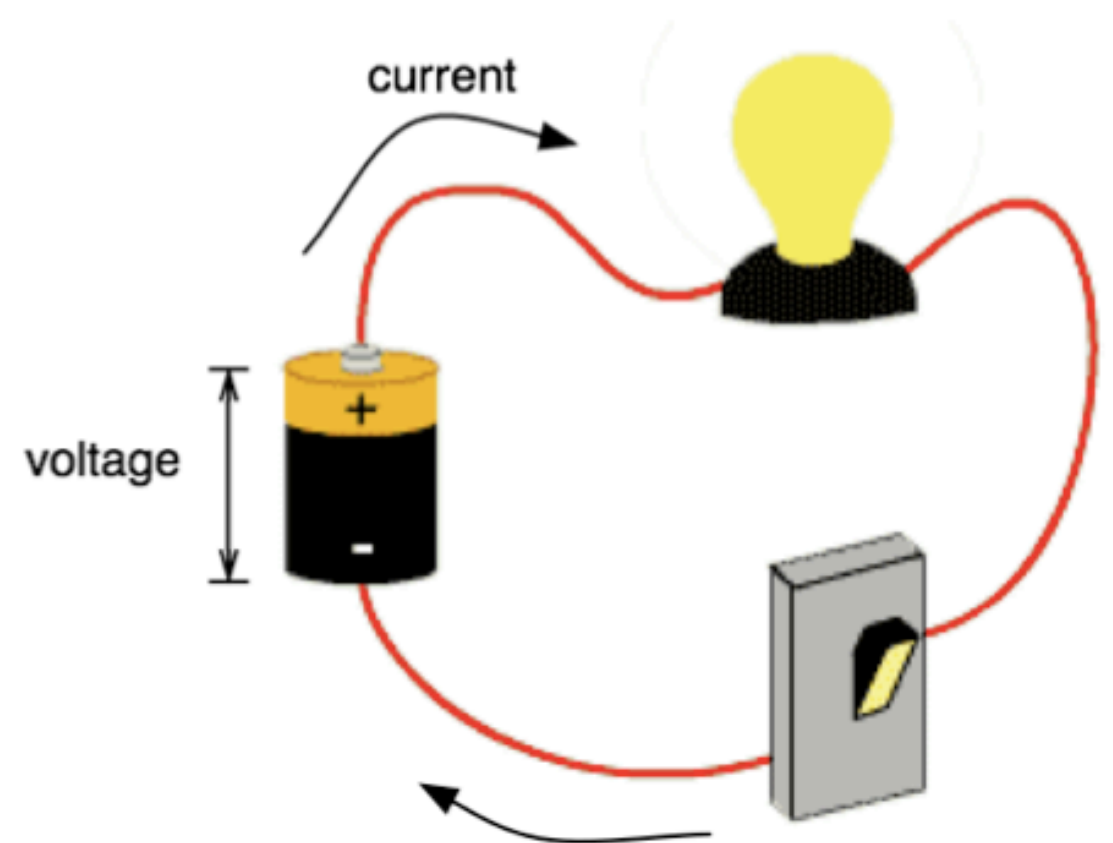
void setup(){
  for (int thisPin=2; thisPin<=8; thisPin++){
    pinMode(thisPin, OUTPUT);
  }

  void loop(){
    // loop from lowest to highest pin
    for (int thisPin=2; thisPin<=8; thisPin++){
      {
        digitalWrite(thisPin, HIGH);
        delay(timer);
        digitalWrite(thisPin, LOW);
      }
    }
    for (int thisPin=8; thisPin>=2; thisPin--){
      digitalWrite(thisPin, HIGH);
      delay(timer);
      digitalWrite(thisPin, LOW);
    }
  }
}
```



Sensors and inputs

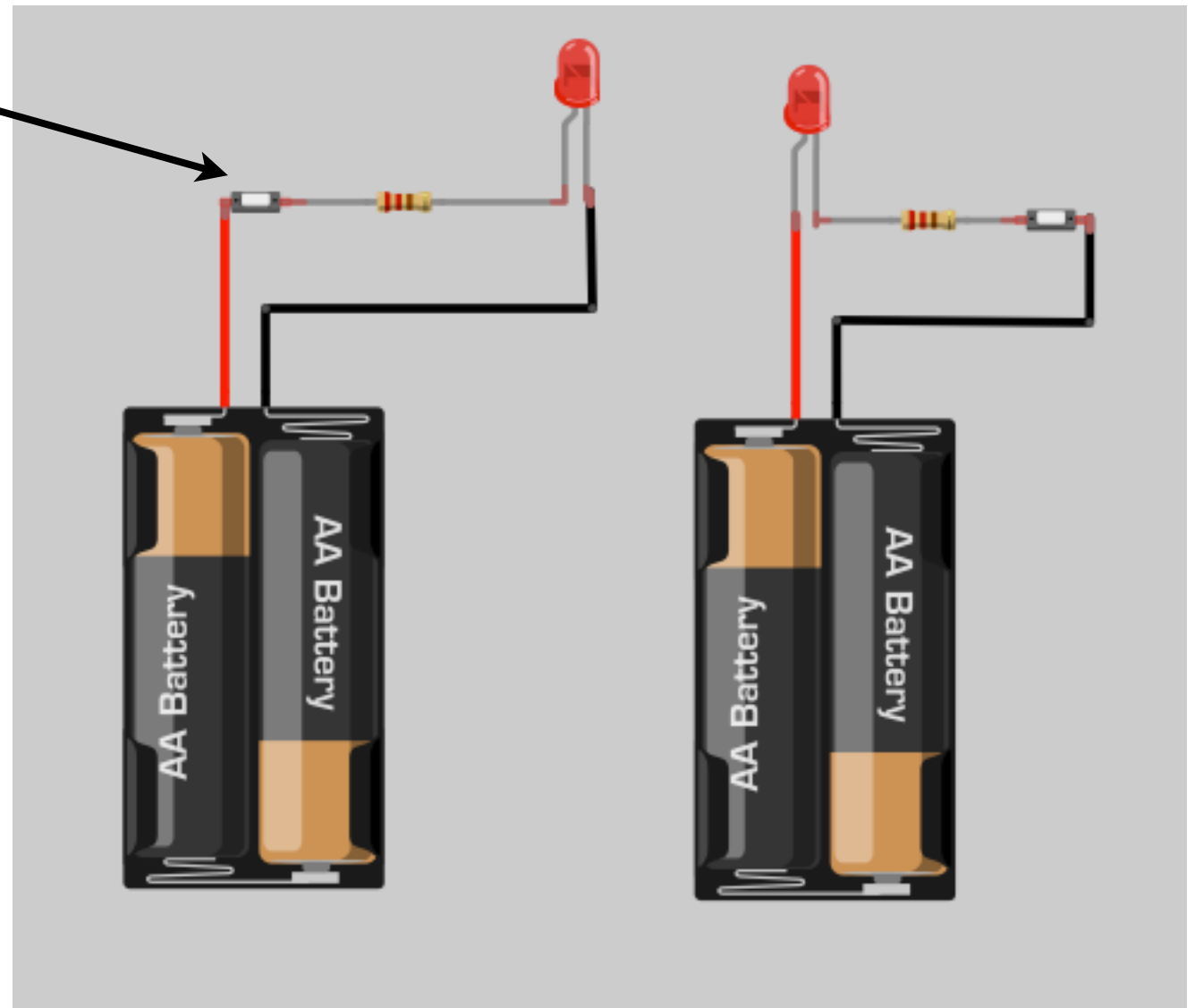
- Many sensors are variations on switches. Switches make or break a connection.
- Button ON:
 - Closed circuit, so current flows
- Button OFF:
 - Open circuit, no current



Push buttons and LEDs

- Once again, there is no difference between these two circuits.

Button

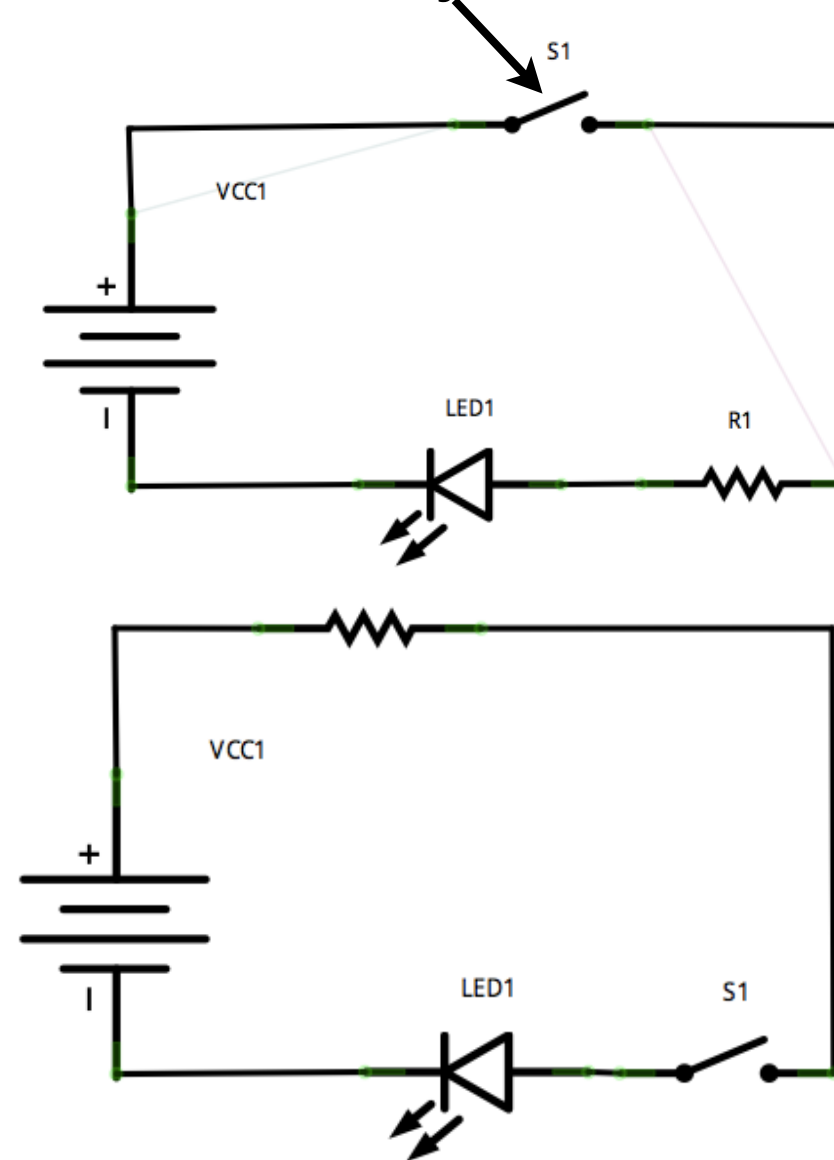


Single pole, single throw (SPST)

What is the difference between these two circuits?



SPST Symbol



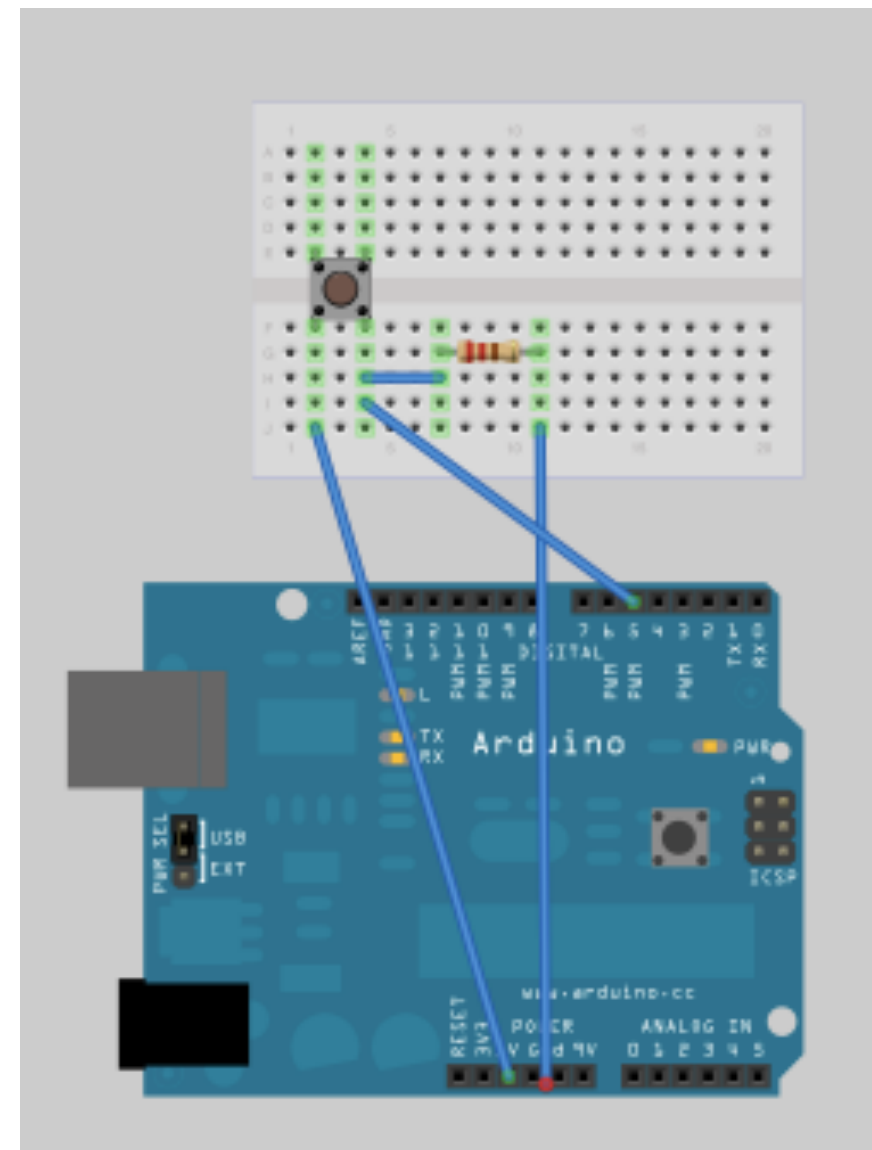


Reading input button

```
int pinUT=5;
int data=0;

void setup()
{
  //pin5 in set to receive data
  pinMode(pinUT, INPUT);
  //initiate serial communication
  Serial.begin(9600);
}

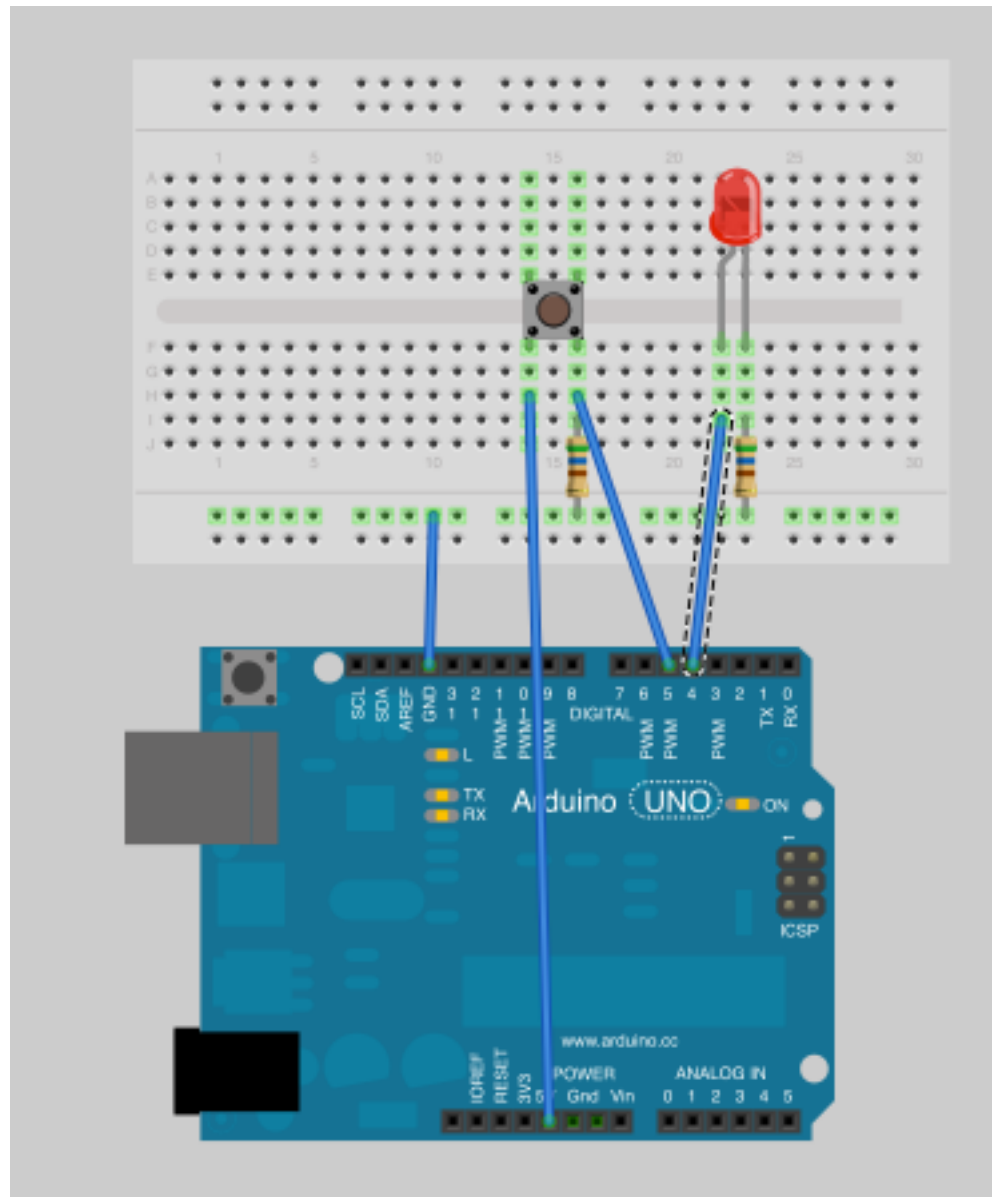
void loop() {
  data=digitalRead(pinUT);
  Serial.println(data);
  delay(1000);
}
```



Serial monitor will report “1” when button is pressed.
It will report “0” otherwise.



Turn ON a LED with a input button



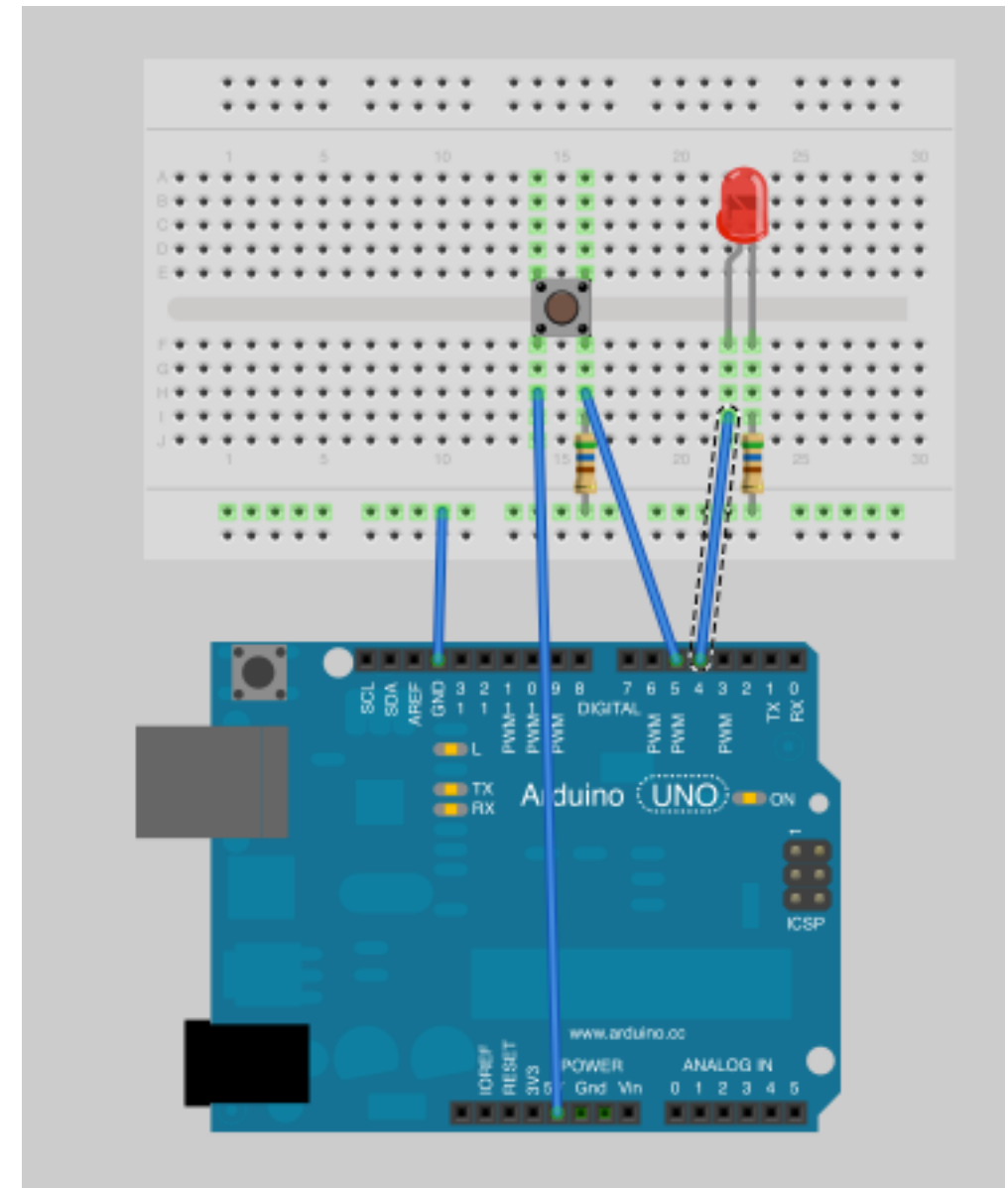
- Turn ON/OFF an LED attached to pin 4 as you press a button.
- Do not get rid of this circuit... we are going to re-use it.

Turn ON a LED with a input button (possible code)

```
int pinUT=5;
int pinLED=4;
int data=0;

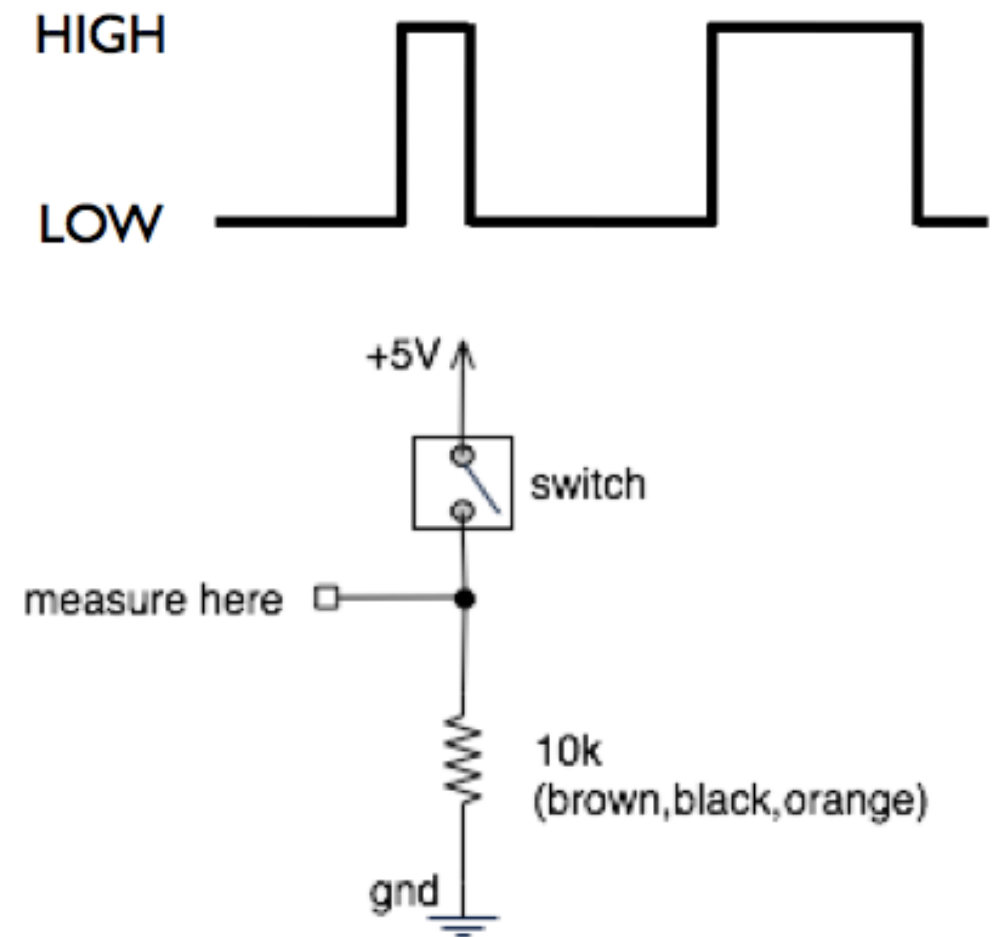
void setup()
{
    //pin #5 in set to receive data
    pinMode(pinUT, INPUT);
    //pin #4 has an LED
    pinMode(pinLED, OUTPUT);
}

void loop() {
    data=digitalRead(pinUT);
    if (data==1 {
        digitalWrite(pinLED,HIGH);
    }
    else digitalWrite(pinLED,LOW);
    delay(1000);
}
```

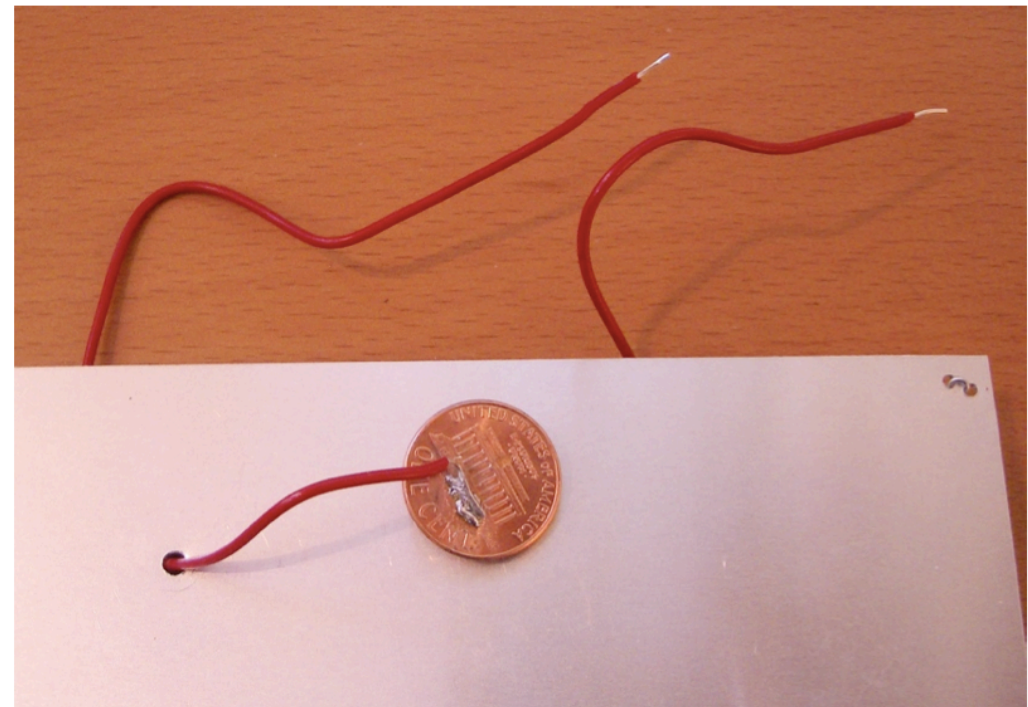


Why do we read button input like that?

- Switches make or break a connection
- But Arduino wants to see a voltage
- Specifically, a “HIGH” (5 volts) or a “LOW” (0 volts)
- How do you go from make/break to HIGH/LOW?



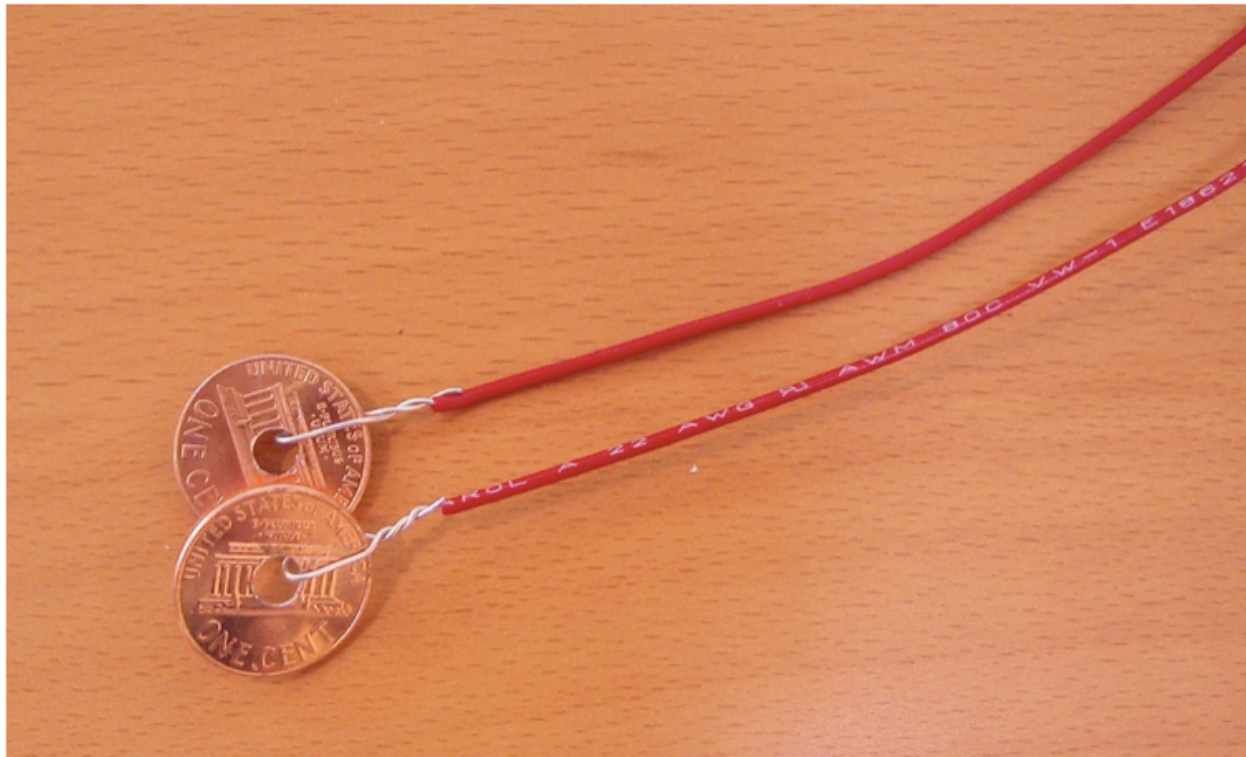
One possible switch



- Switches : “Make” or “Break” an electrical connection
- Eg: Penny on a surface. When the penny is lifted, alarms go off



Another possible switch



- Switches : “Make” or “Break” an electrical connection.
- Eg: When two pennies collide an electrical connection is made.
- Turn ON/OFF a LED on based on the contact of a button.